# How do Firms Behave in Simplified Tax Regimes?

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The views expressed in this presentation are those of the authors and do not necessarily represent those of the Receita Federal do Brasil (RFB). Any analysis of RFB tax data was conducted on RFB's premises by an authorized tax auditor. This work is part of the research collaboration between the ITO and the RFB.

# Outline

1. Motivation

2. Context and Data

3. Intensive Margin Responses

4. Extensive Margin Responses

# Motivation

**Small and medium-sized firms (SMEs) play a central role in economic activity worldwide**

- *How* to tax these firms is a central question
- A common solution is to use **simplified tax regimes** based on **revenue taxation**

$\longrightarrow$ More frequent and larger coverage in low and middle-income countries See

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**The desirability of these special tax regimes for SMEs is ambiguous**

1. On the gains side
   - "Good" compliers
   - Increase registration, reporting, and real output
2. On the losses side
   - "Bad" compliers
   - They would register anyway, and real decisions would not be distorted
   - Weak-monitoring may facilitate avoidance/evasion

**Despite its popularity, very little research on firms' behavior in simplified tax regimes**

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# This Paper

**This paper studies a policy that reduced revenue taxes to foster registration and employment but it also opened loopholes for income shifting**

**Factor R:** It is a payroll-to-revenue ratio

- Firms in eligible activities with Factor R $\geq 0.28$ receive $\sim$ 9pp revenue tax cut
- Introduces a significant notch in the tax schedule
- Total Payroll: Gross wages, owner wages (**prolabore**), contractors, CPP!, FGTS, $13^{th}$ salary

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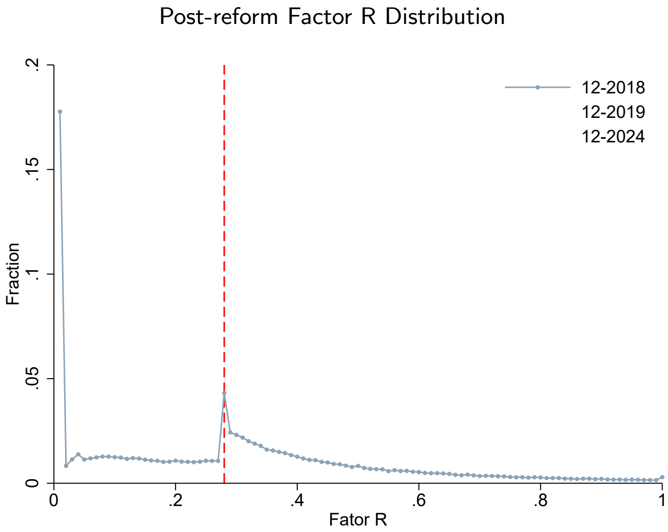
# Preview of the Results



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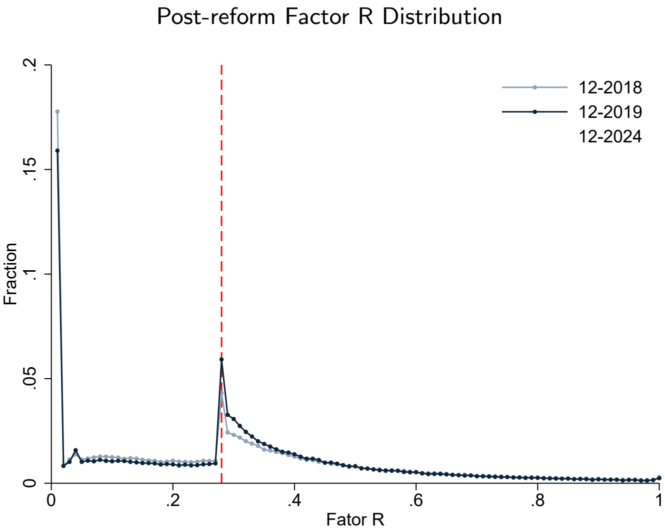

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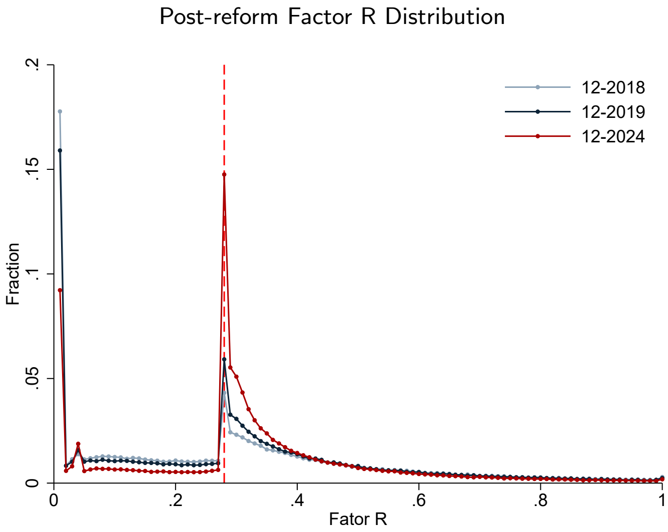


**Bunching grows over time: learning, new firm entry, or both?**

# The Policy

1. We rethink the policy as generating two clean quasi-experiments:
    - Firms with pre-policy FR $\geq .28$ automatically eligible for the tax cut
    - Firms with pre-policy FR $< .28$ need to reach the threshold
2. As the policy targeted economic activities, we can study both:
    - Responses to incumbent firms (intensive margin)
    - Responses to new firms (extensive margin)

# Preview of the Results

1. How do firms respond to a large reduction in the revenue tax rate?
2. How do firms respond to tax evasion/avoidance opportunities?

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1. How do firms respond to a large reduction in the revenue tax rate?
   - Incumbent firms (intensive margin)
     - Sizable reported revenue responses ($\varepsilon = .9$)
     - Null employment effects
   - New firms (extensive margin)
     - Large increase in firms' registration in the simplified tax regime
     - (Preliminary) Little evidence of substituting other tax regimes
2. How do firms respond to tax evasion/avoidance opportunities?

# Preview of the Results

1. How do firms respond to a large reduction in the revenue tax rate?
2. How do firms respond to tax evasion/avoidance opportunities?
   - Incumbent firms (intensive margin )
     - Very large income-shifting response
     - Null revenue response
   - New firms (extensive margin)
     - Compositional analysis suggests entry is driven by income-shifting opportunity

# Outline

# Tax Regimes in Brazil

Totals by Tax Regime (2022)

| | Firms | Tax Payments (in millions R$) | Employees | Payroll (in millions R$) |
|---|---|---|---|---|
| Lucro Real | 220,384 | 1,354,174 | 34,126,577 | 1,584,977 |
| Lucro Presumido | 1,073,033 | 201,994 | 7,745,465 | 290,290 |
| Simples Nacional | 5,560,139 | 150,618 | 19,000,890 | 660,637 |
| Totals | 6,853,556 | 1,706,787 | 60,872,933 | 2,535,905 |
| Share Simples | 81.13% | 8.08% | 31.21% | 26.05% |

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It does not include low-income self-employed (covered by MEI $<$ R$ 81,000 (US$ 16,500)) See

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# Simplified Tax Regime: Industry Classification

- Anexo I: Commerce
- Anexo II: Manufacturing
- Anexo III: Services I
- Anexo IV: Services II (*legal, landscaping, interior design, security, cleaning, maintenance*)
- Anexo V: Services III (*engineering, medicine, auditing, consulting, journalism, advertising*)

**Anexos define the tax rate firms pay**

- Firms in Anexo V are subject to Factor R

# Simplified Tax Regime: Marginal Tax Rates

Tax Schedule by Anexo

| **Annual Revenues (R$)** | **Anexo III** | **Anexo IV** | **Anexo V** |
|---|---|---|---|
| $\leq$ 180,000 | 6.00% | 4.50% | 15.50% |
| 180,000 to 360,000 | 11.20% | 9.00% | 18.00% |
| 360,000 to 720,000 | 13.50% | 10.20% | 19.50% |
| 720,000 to 1,800,000 | 16.00% | 14.00% | 20.50% |
| 1,800,000 to 3,600,000 | 21.00% | 22.00% | 23.00% |
| 3,600,000 to 4,800,000 | 33.00% | 33.00% | 30.50% |

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**Firms in Anexo V are switched to Anexo III if Fator R $\geq$ 0.28** Plots

# The Introduction of the Factor R Rule

In January 2018, the Factor R rule took effect, defining eligibility based on economic activity

1. All activities previously in Anexo VI (most important group) Old tax rates
   - Anexo VI was eliminated (recall: *consulting, engineering, journalism, doctors, etc.*)
   - It serves as the **treated group**
2. Excluded activities: those previously in Anexo IV Old tax rates
   - Not eligible and no incentives to report different activity
   - It serves as the **control group** (recall: *lawyers, landscaping, interior design, etc.*)

# Data

**Tax Data (PGDAS) $\rightarrow$ very simple!**

- Monthly
- Self-reported Anexo (determines tax rate)
- Self-report current and accumulated 12-month revenues
- If subject to Factor R, self-report current and accumulated 12-month total payroll

These firms do not fill any other tax form $\rightarrow$ simple for them, hard for us...

**How can we observe payroll for firms not subjected to Factor R?**

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**Electronic Reporting System (GFIP)**

- Firm owners are mandated to report labor, social security, and tax obligations
- Complex data at the *rubric* level: payment reason

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2. Disadvantage: In practice, it doesn't map 1-to-1 to the payroll reported in tax data

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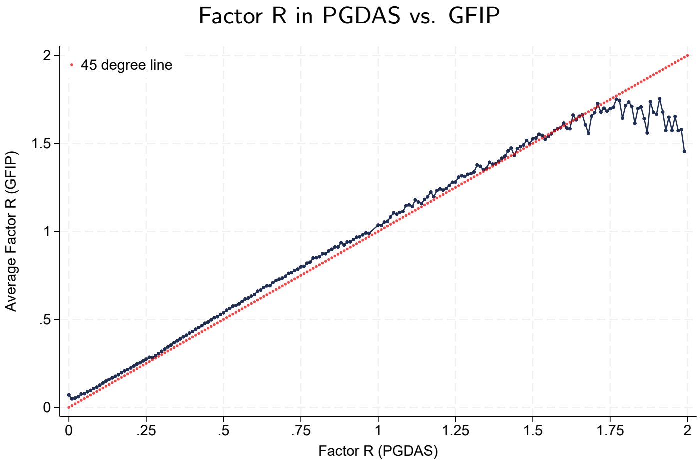
# Comparison of Factor R across different sources




**Factor R constructed from GFIP closely tracks Factor R reported in PGDAS**

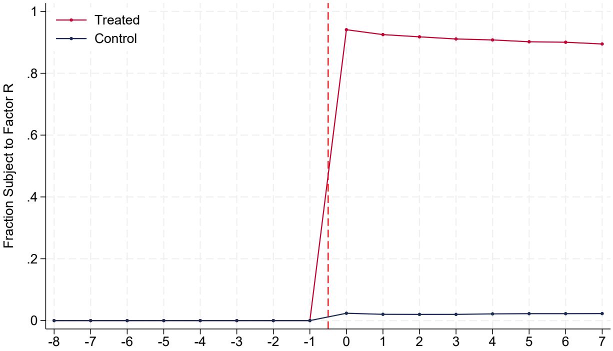
# Outline

# Sample Construction

To keep composition fixed, we follow incumbent firms from 2016 to 2019

- Treatment group: Anexo VI pre-reform
- Control group: Anexo IV pre-reform

Fraction of Firms Subject to Factor R Regime



# Sample Construction

Descriptives Statistics: Firms in 2017-Q4

| | Accumulated 12-month | | Current (Quarter) | |
|---|---|---|---|---|
| | Treated | Control | Treated | Control |
| Revenues | 268,926 | 532,082 | 24,873 | 49,829 |
| Owner Wages | 24,902 | 22,201 | 2,144 | 1,875 |
| Employees' Payroll | 44,833 | 106,803 | 4,668 | 10,969 |
| $\mathbf{1}$[Revenues $= 0$] | 0 | 0 | 0 | 0 |
| $\mathbf{1}$[Owner Wages $= 0$] | 0.194 | 0.265 | 0.211 | 0.288 |
| $\mathbf{1}$[Employees' Payroll $= 0$] | 0.440 | 0.335 | 0.461 | 0.366 |
| $\mathbf{1}$[Total Payroll $= 0$] | 0.100 | 0.138 | 0.110 | 0.154 |
| N Firms | 36,183 | 47,286 | 36,183 | 47,286 |

US\$ 1 $\simeq$ R\$3.34 (Dec - 2017) By FR

# Reduced Form Analysis

We use a simple difference-in-differences design:

$$y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}[\text{Anexo VI} = 1]\mathbf{1}[t = k] + u_{it} \tag{1}$$

- $y_{it}$ is the outcome variable: revenues, owner wages, payroll, etc.
- $\alpha_i$ and $\lambda_t$ are firm and time FE, respectively
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- Standard errors clustered at the industry level

Caveat: We estimate both OLS and Poisson models when necessary

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# Reduced Form Analysis

1. Effects on the entire Factor R distribution
    - *What explains the bunching?*

2. Split firms by pre-policy Factor R
    - Firms with FR $\geq 28\%$ $\rightarrow$ Large revenue tax cut
    - Firms with FR $< 28\%$ $\rightarrow$ Large income-shifting incentives

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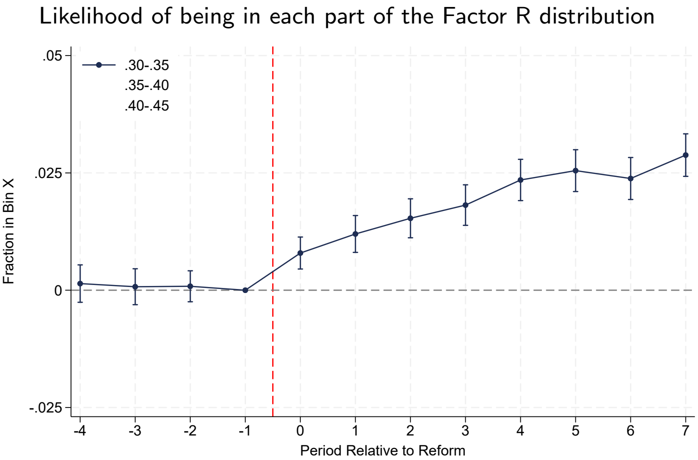
# Effects on the location in the Factor R distribution

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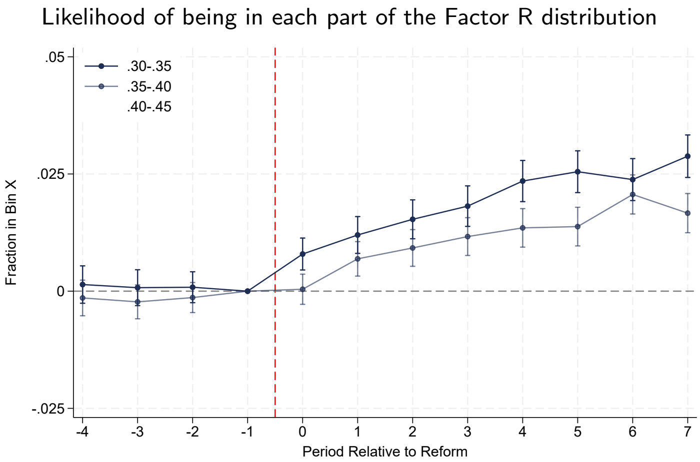

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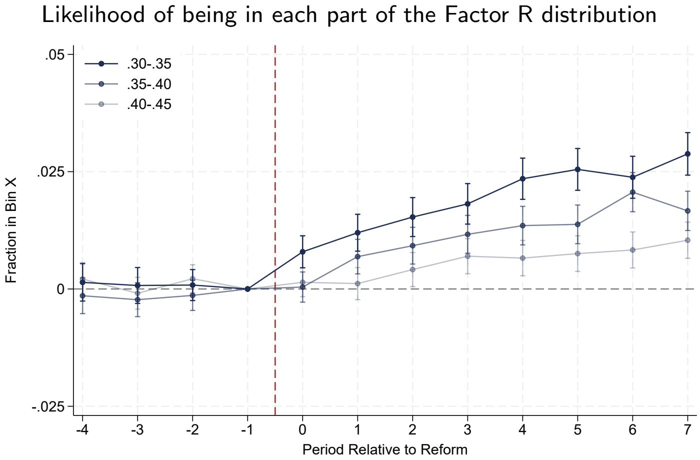

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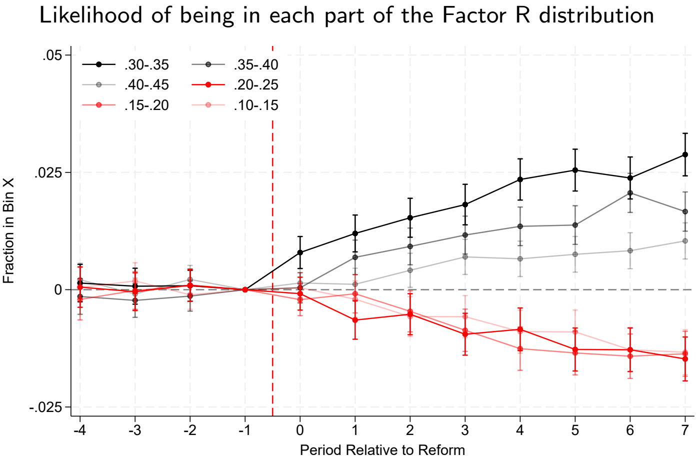
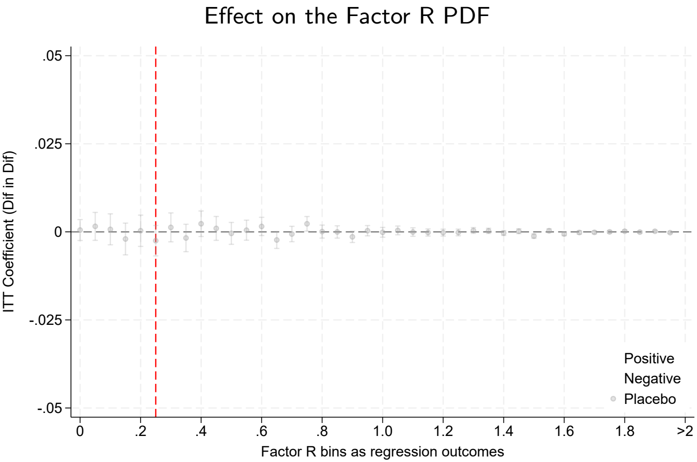



**Increase (decrease) in the likelihood of being in bins above (below) threshold**

# Effects on the location in the Factor R distribution




**Each dot uses a different FR-bin as outcome and estimates the dif-in-dif specification**

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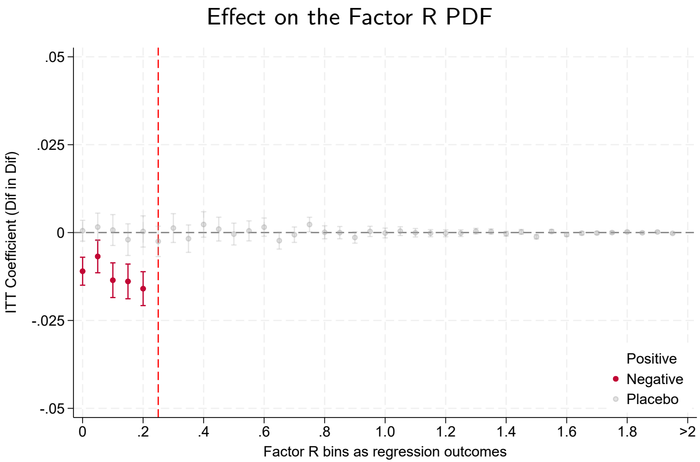

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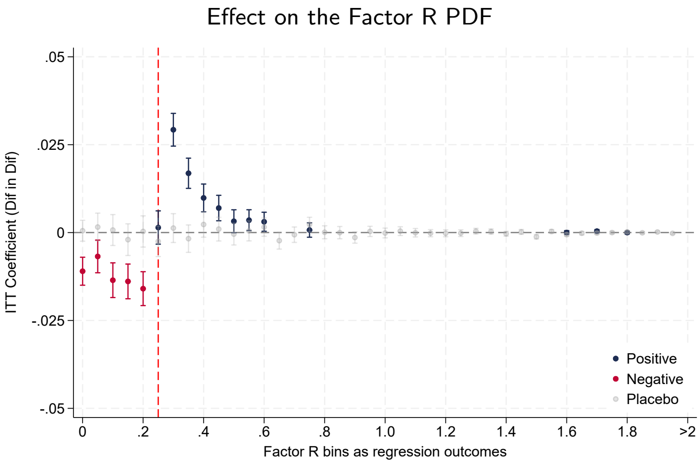

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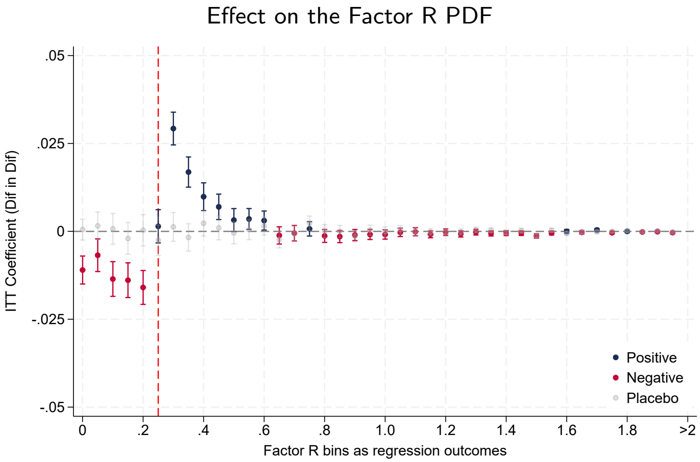
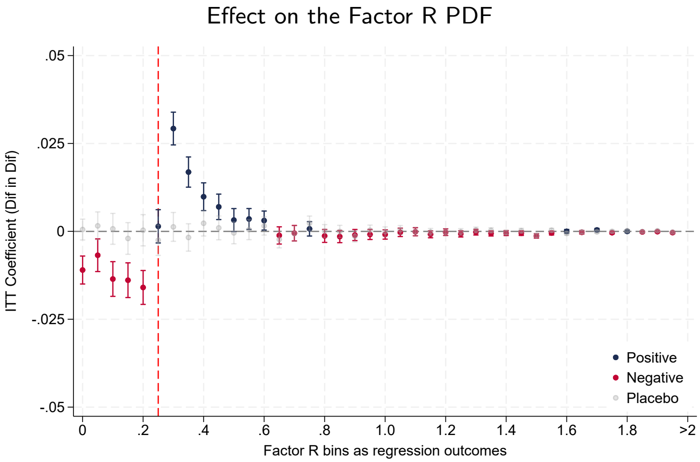



**The increase just above 28% comes from a drop in mass below 28% and above 60%**

# Effects on the location in the Factor R distribution




**84% of the bunching comes from below the threshold, and 16% comes from above** Avg. FR

# Splitting in High and Low Factor R

The policy changes incentives differently depending on firms' pre-policy Factor R

1. High FR firms: were facing $\tau_B \rightarrow$ **automatically eligible** for $\tau_A < \tau_B$
    - Unconstrained case: $\left(\tilde{R}^B, R(e^B, L^B), w_o^B\right) \rightarrow \left(\tilde{R}^A, R(e^A, L^A), w_o^A\right)$ if $\text{FR}^A > .28$
    - Constrained case: $\left(\tilde{R}^B, R(e^B, L^B), w_o^B\right) \rightarrow \left(\tilde{R}^{con}, R(e^{con}, L^{con}), w_o^{con}\right)$ with $\text{FR}^{con} = .28$ if $\text{FR}^A < .28$
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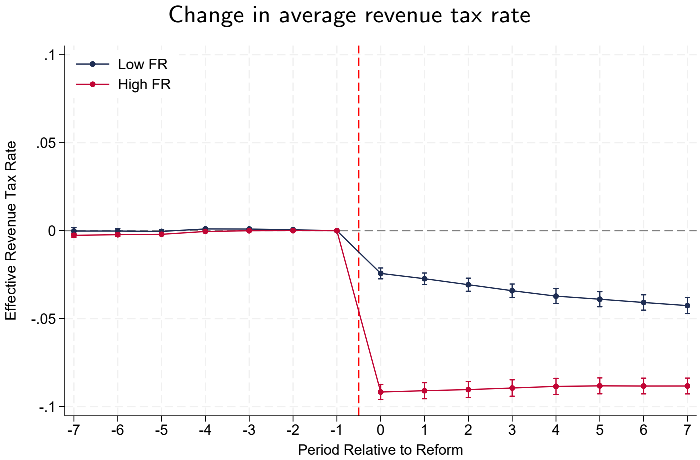
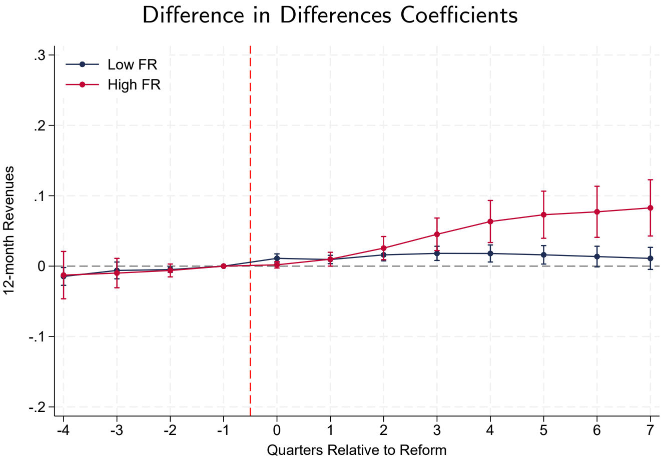
# Effective Tax Rates




**Consistent revenue tax rate response: sharp drop ($>$ .28), smaller and gradual ($<$.28)**

Levels High Levels Low

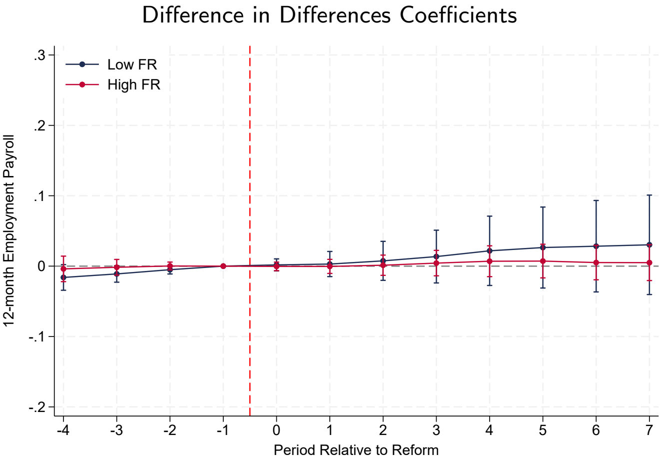
# Revenues




**Reported revenues increase by around 10% among high-Factor-R firms**

Levels High | Levels Low | Current | Heterogeneity | Elasticity

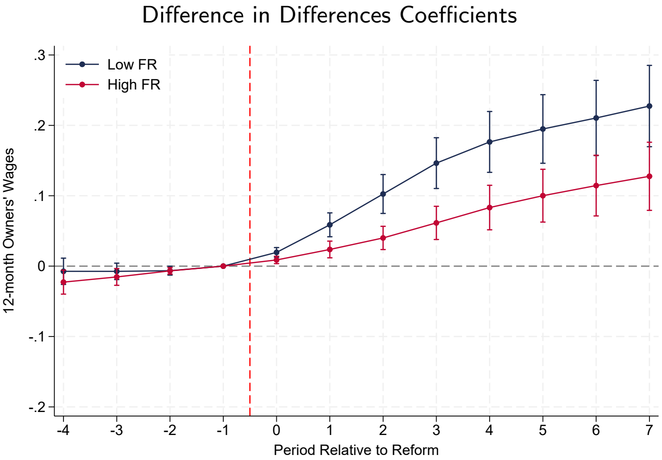
# Employees' Payroll




**Null effect for high Factor R. Small (if any) for low Factor R**

Levels High | Levels Low | Current

# Owner Wages




**Large responses in Owner Wages, especially for low Factor R firms**

Levels High | Levels Low | Current | Heterogeneity | Elasticity | Sh. Wages High | Sh. Wages Low

# Summary of Results for the Intensive Margin

Incumbent firms strongly respond to the policy, relocating close to the threshold

→ Most come from below the threshold, but some also from far above

When splitting the sample by the different incentives generated by the policy

1. High-Factor-R firms
   → Sizable revenue response ($\sim 10\%$), but null effect on employment
   → Firms just above the threshold also increase owner wages to relax the constraint
2. Low-Factor-R firms
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   → Increase in owner wages ($\sim 30\%$) with no change in total revenues

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# Extensive Margin

**New firm entry**

1. Industry-level analysis
   - Map industry codes to pre-reform Anexos $\rightarrow$ exposed industries linked to Anexo VI See
   - Compare the total number of new firms per year for exposed vs. non-exposed industries
   - Track changes in the composition of new entrants

2. Discussion: Firm creation?
   - Relabeling industry code ✓
   - Switching tax regime (MEI / Lucro Presumido), labor contract (PJs), registration status ×

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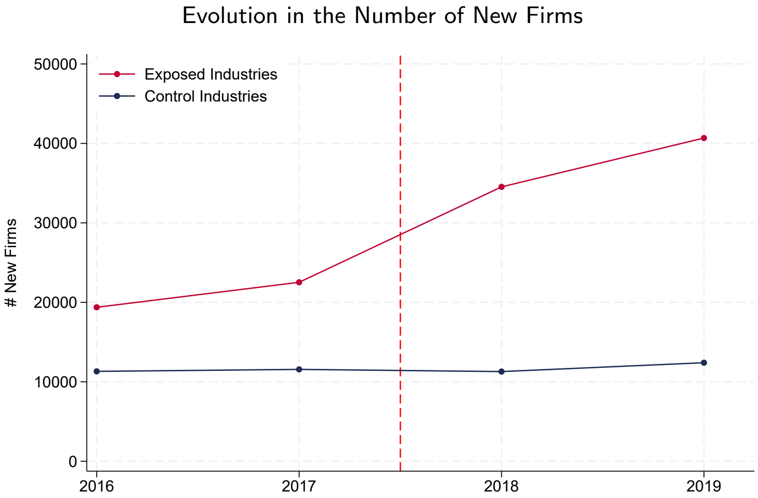
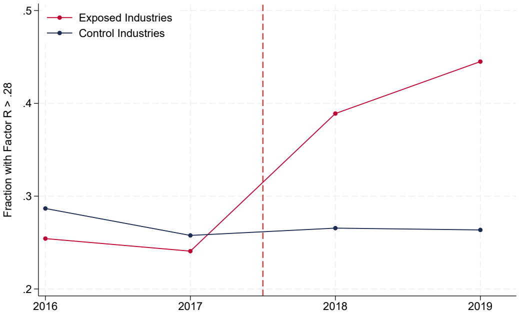



**The number of new firms increases substantially in exposed industries**

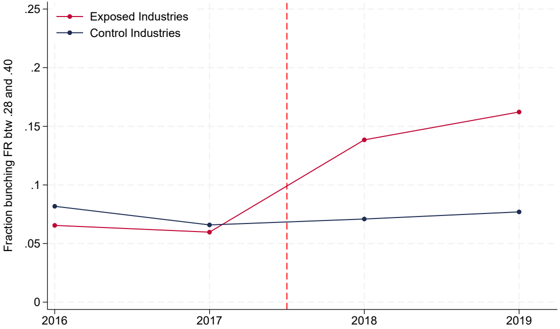
Relabeling

# Composition of New Firms

Factor R of the Average New Firm




(a) FR $\geq .28$

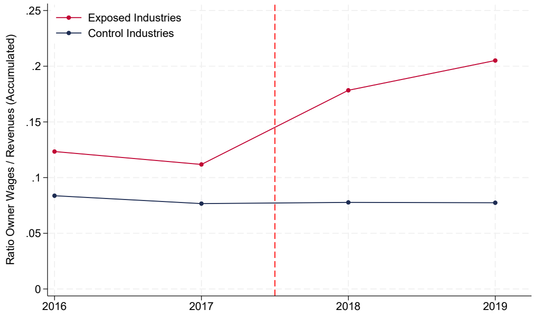



(b) FR $\in [.28, .40]$

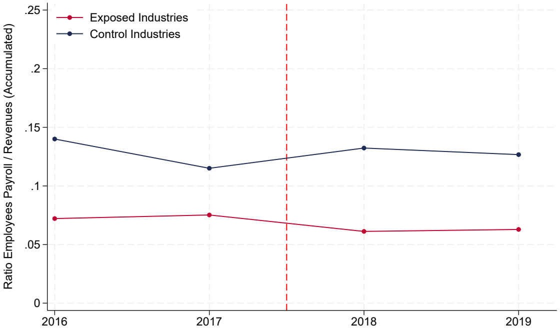
**After the reform, new firms in exposed industries enter with higher Factor R**

# Composition of New Firms

Decomposing the Total Payroll




(a) Owner Wages / Revenues




(b) Employees Payroll / Revenues

**After the reform, new entrants are not more labor intensive, by behave as income-shifters**

# Summary of Results for the Extensive Margin

We find a large increase in the registration of firms due to a more generous simplified tax regime

1. Little evidence of industry relabeling, but it could be:
    - Substitution of entering into another tax regime
    - Pre-existing but fully informal activities
2. While the policy aimed to help labor-intensive firms:
    - New firms seem to be registering to perform income-shifting
    - Suggest that firms that select into these tax regimes take advantage of the system

Thank you

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# References I


Heim, Bradley T (2010). "The responsiveness of self-employment income to tax rate changes". In: *Labour Economics* 17.6, pp. 940–950.

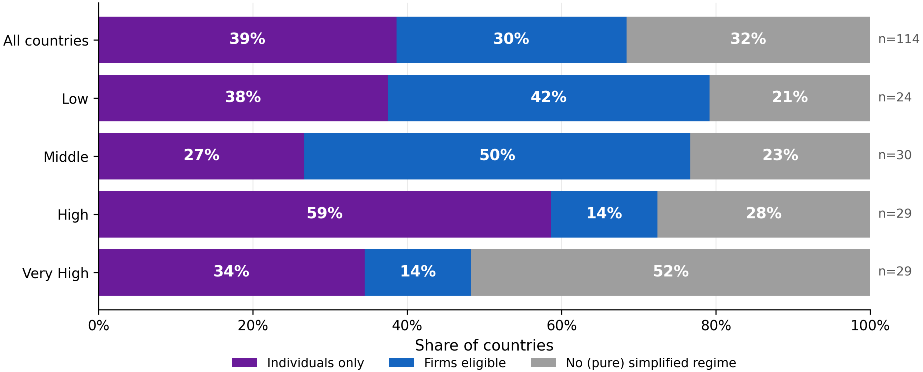
Lobel, Felipe, Thiago Scot, and Pedro Zúniga (2024). "Corporate taxation and evasion responses: Evidence from a minimum tax in Honduras". In: *American Economic Journal: Economic Policy* 16.1, pp. 482–517.

Palomo, Theo, Davi Bhering, Thiago Scot, Pierre Bachas, Luciana Barcarolo, Celso Campos, Javier Feinmann, Leonardo Moreira, and Gabriel Zucman (2025). *Tax Progressivity and Inequality in Brazil: Evidence from Integrated Administrative Data*. Tech. rep. EU Tax Observatory.

# Appendix

# Motivation

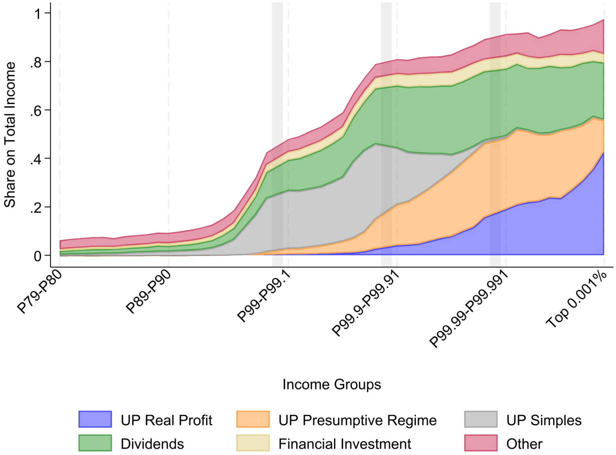
Eligibility criteria and usage of presumptive taxation




Sample: 114 countries. Income groups by GDP per capita current USD (NY.GDP.PCAP.CD): Low ≤ 4, 495; *Middle*4,496–13, 935; *High*13,936–36, 000; *VeryHigh* > 36,000. $4,495 and $13,935 are World Bank FY2026 Low/Lower-middle and Upper-middle ceilings (GNI Atlas, 2024); $36,000 splits WB high-income at its sample median (29/29). Classification uses ONLY pure presumptive regimes; mixed (rate-concession) regimes are ignored. Source: coded_analysis_v3.xlsx, World Bank WDI.

**More frequent and larger coverage in low and middle-income countries** Back

# Who are Simples Nacional Owners?




Source: Palomo et al. 2025 Back

# Tax rates for each Anexo

Table: Anexo I: Commerce

| | Receita Bruta em 12 Meses (em R$) | Alíquota | Valor a Deduzir (em R$) |
|---|---|---|---|
| 1ª Faixa | Até 180.000,00 | 4,00% | - |
| 2ª Faixa | De 180.000,01 a 360.000,00 | 7,30% | 5.940,00 |
| 3ª Faixa | De 360.000,01 a 720.000,00 | 9,50% | 13.860,00 |
| 4ª Faixa | De 720.000,01 a 1.800.000,00 | 10,70% | 22.500,00 |
| 5ª Faixa | De 1.800.000,01 a 3.600.000,00 | 14,30% | 87.300,00 |
| 6ª Faixa | De 3.600.000,01 a 4.800.000,00 | 19,00% | 378.000,00 |

# Tax rates for each Anexo

Table: Anexo II: Manufacturing

| | **Receita Bruta em 12 Meses (em R$)** | **Alíquota** | **Valor a Deduzir (em R$)** |
|---|---|---|---|
| 1ª Faixa | Até 180.000,00 | 4,50% | - |
| 2ª Faixa | De 180.000,01 a 360.000,00 | 7,80% | 5.940,00 |
| 3ª Faixa | De 360.000,01 a 720.000,00 | 10,00% | 13.860,00 |
| 4ª Faixa | De 720.000,01 a 1.800.000,00 | 11,20% | 22.500,00 |
| 5ª Faixa | De 1.800.000,01 a 3.600.000,00 | 14,70% | 85.500,00 |
| 6ª Faixa | De 3.600.000,01 a 4.800.000,00 | 30,00% | 720.000,00 |

# Tax rates for each Anexo

Table: Anexo III: serviços não relacionados no 5o-C do art. 18 desta Lei Complementar

| | **Receita Bruta em 12 Meses (em R$)** | **Alíquota** | **Valor a Deduzir (em R$)** |
|---|---|---|---|
| 1ª Faixa | Até 180.000,00 | 6,00% | – |
| 2ª Faixa | De 180.000,01 a 360.000,00 | 11,20% | 9.360,00 |
| 3ª Faixa | De 360.000,01 a 720.000,00 | 13,50% | 17.640,00 |
| 4ª Faixa | De 720.000,01 a 1.800.000,00 | 16,00% | 35.640,00 |
| 5ª Faixa | De 1.800.000,01 a 3.600.000,00 | 21,00% | 125.640,00 |
| 6ª Faixa | De 3.600.000,01 a 4.800.000,00 | 33,00% | 648.000,00 |

# Tax rates for each Anexo

Table: Anexo IV: serviços relacionados no 5o-C do art. 18 desta Lei Complementar

| | **Receita Bruta em 12 Meses (em R$)** | **Alíquota** | **Valor a Deduzir (em R$)** |
|---|---|---|---|
| 1ª Faixa | Até 180.000,00 | 4,50% | – |
| 2ª Faixa | De 180.000,01 a 360.000,00 | 9,00% | 8.100,00 |
| 3ª Faixa | De 360.000,01 a 720.000,00 | 10,20% | 12.420,00 |
| 4ª Faixa | De 720.000,01 a 1.800.000,00 | 14,00% | 39.780,00 |
| 5ª Faixa | De 1.800.000,01 a 3.600.000,00 | 22,00% | 183.780,00 |
| 6ª Faixa | De 3.600.000,01 a 4.800.000,00 | 33,00% | 828.000,00 |

# Tax rates for each Anexo

Table: Anexo V: serviços relacionados no 5o-I do art. 18 desta Lei Complementar

| | Receita Bruta em 12 Meses (em R$) | Alíquota | Valor a Deduzir (em R$) |
|---|---|---|---|
| 1ª Faixa | Até 180.000,00 | 15,50% | – |
| 2ª Faixa | De 180.000,01 a 360.000,00 | 18,00% | 4.500,00 |
| 3ª Faixa | De 360.000,01 a 720.000,00 | 19,50% | 9.900,00 |
| 4ª Faixa | De 720.000,01 a 1.800.000,00 | 20,50% | 17.100,00 |
| 5ª Faixa | De 1.800.000,01 a 3.600.000,00 | 23,00% | 62.100,00 |
| 6ª Faixa | De 3.600.000,01 a 4.800.000,00 | 30,50% | 540.000,00 |

# Before the reform

Table: Anexo VI: prestação de serviços relacionados no 5o-I do art. 18 desta Lei Complementar

| Receita Bruta em 12 meses (em R$) | Alíquota | IRPJ, PIS/Pasep, CSLL, Cofins e CPP | ISS |
|---|---|---|---|
| Até 180.000,00 | 16,93% | 14,93% | 2,00% |
| De 180.000,01 a 360.000,00 | 17,72% | 14,93% | 2,79% |
| De 360.000,01 a 540.000,00 | 18,43% | 14,93% | 3,50% |
| De 540.000,01 a 720.000,00 | 18,77% | 14,93% | 3,84% |
| De 720.000,01 a 900.000,00 | 19,04% | 15,17% | 3,87% |
| De 900.000,01 a 1.080.000,00 | 19,94% | 15,71% | 4,23% |
| De 1.080.000,01 a 1.260.000,00 | 20,34% | 16,08% | 4,26% |
| De 1.260.000,01 a 1.440.000,00 | 20,66% | 16,35% | 4,31% |
| De 1.440.000,01 a 1.620.000,00 | 21,17% | 16,56% | 4,61% |
| De 1.620.000,01 a 1.800.000,00 | 21,38% | 16,73% | 4,65% |
| De 1.800.000,01 a 1.980.000,00 | 21,86% | 16,86% | 5,00% |
| De 1.980.000,01 a 2.160.000,00 | 21,97% | 16,97% | 5,00% |
| De 2.160.000,01 a 2.340.000,00 | 22,06% | 17,06% | 5,00% |
| De 2.340.000,01 a 2.520.000,00 | 22,14% | 17,14% | 5,00% |
| De 2.520.000,01 a 2.700.000,00 | 22,21% | 17,21% | 5,00% |
| De 2.700.000,01 a 2.880.000,00 | 22,21% | 17,21% | 5,00% |
| De 2.880.000,01 a 3.060.000,00 | 22,32% | 17,32% | 5,00% |
| De 3.060.000,01 a 3.240.000,00 | 22,37% | 17,37% | 5,00% |
| De 3.240.000,01 a 3.420.000,00 | 22,41% | 17,41% | 5,00% |
| De 3.420.000,01 a 3.600.000,00 | 22,45% | 17,45% | 5,00% |

**Independent of Factor R** Back

# Before the Reform: Anexo V

**Anexo V** applied to serviços relacionados no **5o-D** do art. 18 desta Lei Complementar

- These firms were moved to Anexo III
- They had a tax schedule dependent on a complex Factor R Back Distribution

| **RB em 12 meses** | $(r) < 0{,}10$ | 0,10 – 0,15 | 0,15 – 0,20 | 0,20 – 0,25 | 0,25 – 0,30 | 0,30 – 0,35 | 0,35 – 0,40 | $(r) \geq 0{,}40$ |
|---|---|---|---|---|---|---|---|---|
| Até 180k | 17,50% | 15,70% | 13,70% | 11,82% | 10,47% | 9,97% | 9,10% | 8,00% |
| 180k a 360k | 17,52% | 15,75% | 13,90% | 12,60% | 12,33% | 10,72% | 9,10% | 8,48% |
| 360k a 540k | 17,55% | 15,95% | 14,20% | 12,90% | 12,64% | 11,14% | 9,58% | 9,03% |
| 540k a 720k | 17,95% | 16,70% | 15,00% | 13,70% | 13,45% | 12,00% | 10,56% | 9,34% |
| 720k a 900k | 18,15% | 16,95% | 15,30% | 14,03% | 13,53% | 12,40% | 11,04% | 10,06% |
| 900k a 1.080k | 18,45% | 17,20% | 15,40% | 14,10% | 13,60% | 12,60% | 11,60% | 10,60% |
| 1.080k a 1.260k | 18,55% | 17,30% | 15,50% | 14,11% | 13,68% | 12,69% | 11,69% | 10,69% |
| 1.260k a 1.440k | 18,62% | 17,32% | 15,60% | 14,12% | 13,69% | 12,69% | 11,69% | 10,69% |
| 1.440k a 1.620k | 18,72% | 17,42% | 15,70% | 14,13% | 14,08% | 13,09% | 12,08% | 11,08% |
| 1.620k a 1.800k | 18,86% | 17,56% | 15,80% | 14,14% | 14,09% | 13,09% | 12,09% | 11,09% |
| 1.800k a 1.980k | 18,96% | 17,66% | 15,90% | 14,49% | 14,45% | 13,61% | 12,78% | 11,87% |
| 1.980k a 2.160k | 19,06% | 17,76% | 16,00% | 14,67% | 14,64% | 13,79% | 13,15% | 12,28% |
| 2.160k a 2.340k | 19,26% | 17,96% | 16,20% | 14,86% | 14,82% | 14,17% | 13,54% | 12,68% |
| 2.340k a 2.520k | 19,56% | 18,30% | 16,50% | 15,46% | 15,18% | 14,61% | 14,04% | 13,26% |
| 2.520k a 2.700k | 20,79% | 19,30% | 17,45% | 16,24% | 16,00% | 15,52% | 15,03% | 14,29% |
| 2.700k a 2.880k | 21,20% | 20,00% | 18,28% | 16,91% | 16,72% | 16,32% | 15,93% | 15,23% |
| 2.880k a 3.060k | 21,70% | 20,50% | 18,70% | 17,40% | 17,13% | 16,82% | 16,38% | 16,17% |
| 3.060k a 3.240k | 22,20% | 21,00% | 19,10% | 17,90% | 17,55% | 17,22% | 16,82% | 16,51% |
| 3.240k a 3.420k | 22,50% | 21,30% | 19,50% | 18,20% | 17,97% | 17,44% | 17,24% | 16,94% |
| 3.420k a 3.600k | 22,90% | 21,80% | 20,00% | 18,60% | 18,40% | 17,85% | 17,60% | 17,18% |

# Before the reform

Table: Anexo IV pre-reform

| Receita Bruta em 12 meses (em R$) | Alíquota |
|---|---|
| Até 180.000,00 | 4,50% |
| De 180.000,01 a 360.000,00 | 6,54% |
| De 360.000,01 a 540.000,00 | 7,70% |
| De 540.000,01 a 720.000,00 | 8,49% |
| De 720.000,01 a 900.000,00 | 8,97% |
| De 900.000,01 a 1.080.000,00 | 9,78% |
| De 1.080.000,01 a 1.260.000,00 | 10,26% |
| De 1.260.000,01 a 1.440.000,00 | 10,76% |
| De 1.440.000,01 a 1.620.000,00 | 11,51% |
| De 1.620.000,01 a 1.800.000,00 | 12,00% |
| De 1.800.000,01 a 1.980.000,00 | 12,80% |
| De 1.980.000,01 a 2.160.000,00 | 13,25% |
| De 2.160.000,01 a 2.340.000,00 | 13,70% |
| De 2.340.000,01 a 2.520.000,00 | 14,15% |
| De 2.520.000,01 a 2.700.000,00 | 14,60% |
| De 2.700.000,01 a 2.880.000,00 | 15,05% |
| De 2.880.000,01 a 3.060.000,00 | 15,50% |
| De 3.060.000,01 a 3.240.000,00 | 15,95% |
| De 3.240.000,01 a 3.420.000,00 | 16,40% |
| De 3.420.000,01 a 3.600.000,00 | 16,85% |

Table: Anexo IV post reform

| Receita Bruta em 12 Meses (em R$) | Alíquota |
|---|---|
| Até 180.000,00 | 4,50% |
| De 180.000,01 a 360.000,00 | 9,00% |
| De 360.000,01 a 720.000,00 | 10,20% |
| De 720.000,01 a 1.800.000,00 | 14,00% |
| De 1.800.000,01 a 3.600.000,00 | 22,00% |
| De 3.600.000,01 a 4.800.000,00 | 33,00% |

Back

# Before the reform

Table: Anexo III pre-reform (didn't include 5o-D)

| **Receita Bruta em 12 meses (em R$)** | **ALÍQUOTA** |
|---|---|
| Até 180.000,00 | 6,00% |
| De 180.000,01 a 360.000,00 | 8,21% |
| De 360.000,01 a 540.000,00 | 10,26% |
| De 540.000,01 a 720.000,00 | 11,31% |
| De 720.000,01 a 900.000,00 | 11,40% |
| De 900.000,01 a 1.080.000,00 | 12,42% |
| De 1.080.000,01 a 1.260.000,00 | 12,54% |
| De 1.260.000,01 a 1.440.000,00 | 12,68% |
| De 1.440.000,01 a 1.620.000,00 | 13,55% |
| De 1.620.000,01 a 1.800.000,00 | 13,68% |
| De 1.800.000,01 a 1.980.000,00 | 14,93% |
| De 1.980.000,01 a 2.160.000,00 | 15,06% |
| De 2.160.000,01 a 2.340.000,00 | 15,20% |
| De 2.340.000,01 a 2.520.000,00 | 15,35% |
| De 2.520.000,01 a 2.700.000,00 | 15,48% |
| De 2.700.000,01 a 2.880.000,00 | 16,85% |
| De 2.880.000,01 a 3.060.000,00 | 16,98% |
| De 3.060.000,01 a 3.240.000,00 | 17,13% |
| De 3.240.000,01 a 3.420.000,00 | 17,27% |
| De 3.420.000,01 a 3.600.000,00 | 17,42% |

# Before the reform

Table: Anexo II pre-reform

| | |
|---|---|
| Até 180.000,00 | 4,50% |
| De 180.000,01 a 360.000,00 | 5,97% |
| De 360.000,01 a 540.000,00 | 7,34% |
| De 540.000,01 a 720.000,00 | 8,04% |
| De 720.000,01 a 900.000,00 | 8,10% |
| De 900.000,01 a 1.080.000,00 | 8,78% |
| De 1.080.000,01 a 1.260.000,00 | 8,86% |
| De 1.260.000,01 a 1.440.000,00 | 8,95% |
| De 1.440.000,01 a 1.620.000,00 | 9,53% |
| De 1.620.000,01 a 1.800.000,00 | 9,62% |
| De 1.800.000,01 a 1.980.000,00 | 10,45% |
| De 1.980.000,01 a 2.160.000,00 | 10,54% |
| De 2.160.000,01 a 2.340.000,00 | 10,63% |
| De 2.340.000,01 a 2.520.000,00 | 10,73% |
| De 2.520.000,01 a 2.700.000,00 | 10,82% |
| De 2.700.000,01 a 2.880.000,00 | 11,73% |
| De 2.880.000,01 a 3.060.000,00 | 11,82% |
| De 3.060.000,01 a 3.240.000,00 | 11,92% |
| De 3.240.000,01 a 3.420.000,00 | 12,01% |
| De 3.420.000,01 a 3.600.000,00 | 12,11% |

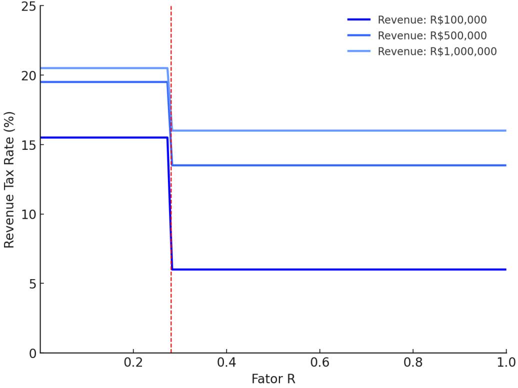
# Before the reform

Table: Anexo I pre-reform

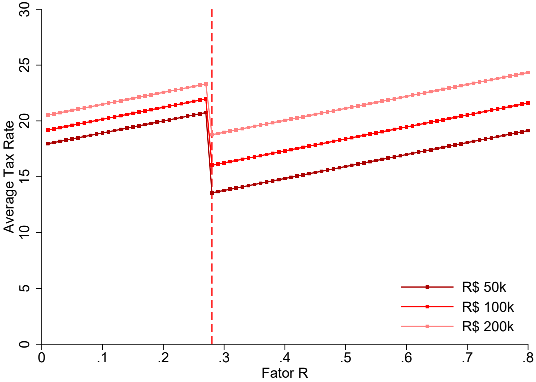
| | |
|---|---|
| Até 180.000,00 | 4,00% |
| De 180.000,01 a 360.000,00 | 5,47% |
| De 360.000,01 a 540.000,00 | 6,84% |
| De 540.000,01 a 720.000,00 | 7,54% |
| De 720.000,01 a 900.000,00 | 7,60% |
| De 900.000,01 a 1.080.000,00 | 8,28% |
| De 1.080.000,01 a 1.260.000,00 | 8,36% |
| De 1.260.000,01 a 1.440.000,00 | 8,45% |
| De 1.440.000,01 a 1.620.000,00 | 9,03% |
| De 1.620.000,01 a 1.800.000,00 | 9,12% |
| De 1.800.000,01 a 1.980.000,00 | 9,95% |
| De 1.980.000,01 a 2.160.000,00 | 10,04% |
| De 2.160.000,01 a 2.340.000,00 | 10,13% |
| De 2.340.000,01 a 2.520.000,00 | 10,23% |
| De 2.520.000,01 a 2.700.000,00 | 10,32% |
| De 2.700.000,01 a 2.880.000,00 | 11,23% |
| De 2.880.000,01 a 3.060.000,00 | 11,32% |
| De 3.060.000,01 a 3.240.000,00 | 11,42% |
| De 3.240.000,01 a 3.420.000,00 | 11,51% |
| De 3.420.000,01 a 3.600.000,00 | 11,61% |

# Tax Notch based on Factor R

Average Tax Rate as a function of Factor R




(a) Not including payroll taxes




(b) Including payroll taxes

Back

Table: LC 123, art. 18, § 5º-I —Subject to Anexo VI (Pre-reform)

| Item | Atividade |
|---|---|
| I | medicina, inclusive laboratorial e enfermagem |
| II | medicina veterinária |
| III | odontologia |
| IV | psicologia, psicanálise, terapia ocupacional, acupuntura, podologia, fonoaudiologia, clínicas de nutrição e de vacinação e bancos de leite |
| V | serviços de comissaria, de despachantes, de tradução e de interpretação |
| VI | arquitetura, engenharia, medição, cartografia, topografia, geologia, geodésia, testes, suporte e análises técnicas e tecnológicas, pesquisa, design, desenho e agronomia |
| VII | representação comercial e demais atividades de intermediação de negócios e serviços de terceiros |
| VIII | perícia, leilão e avaliação |
| IX | auditoria, economia, consultoria, gestão, organização, controle e administração |
| X | jornalismo e publicidade |
| XI | agenciamento, exceto de mão de obra |
| XII | outras atividades do setor de serviços decorrentes do exercício de atividade intelectual, de natureza técnica, científica, desportiva, artística ou cultural, que constitua profissão regulamentada ou não, desde que não sujeitas à tributação na forma dos Anexos III, IV ou V |

Back

# Incentives under Factor R

- Large drop in the revenue tax rate for Factor R $> 0.28$ (even accounting for SSC and IRPF)
- Tax notch $\rightarrow$ drop in the average tax rate between 6 to 9pp

Example for an owner-manager making R$ 350,000, per year (US$ 5,630 monthly, top 1%)

1. All as revenues $\rightarrow$ 0.18 x (350k - 180k) + 0.155 x 180k $\rightarrow$ Av. $\tau$ = **16.7%**
2. 28% as own wages (payroll) = R$ 98,000
   - (0.28 x 350k)x 0.11 **SSC** + ((0.28 x 350k)x 0.89)x 0.13 **IRPF** = 10.8k + 11.3k = 22.1k
   - 0.112 x (350k - 180k) + 0.06 x 180k = 29.84k (av. revenue $\tau$ = 8.5%)
   - Total av. tax rate (incl. SSC) $\tau$ = **15%**

# Incentives under Factor R

- Large drop in the revenue tax rate for Factor R $> 0.28$ (even accounting for SSC and IRPF)
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Example for an owner-manager making R$ 350,000, per year (US$ 5,630 monthly, top 1%)

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2. 28% as own wages (payroll) = R$ 98,000
   - (0.28 x 350k)x 0.11 **SSC** + ((0.28 x 350k)x 0.89)x 0.13 **IRPF** = 10.8k + 11.3k = 22.1k
   - 0.112 x (350k - 180k) + 0.06 x 180k = 29.84k (av. revenue $\tau$ = 8.5%)
   - Total av. tax rate (incl. SSC) $\tau$ **= 15%**

# Outcome Variables

Outcomes in our setting are non-negative and frequently equal to zero (payroll, prolabore, etc.)

- Log-linear models drop zeros, which is an important margin of response
- Log-linear fixes (e.g. $log(Y+1)$) are not unit-invariant
- We use a Poisson Pseudo Maximum Likelihood (PPML)

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -6} \beta_k \mathbf{1}\{\text{Anexo} = 6\} \cdot \mathbf{1}\{t - 201712 = k\} + \theta' X_i + \varepsilon_{it}, \qquad (2)$$

$$\mathbb{E}[Y_{it} \mid D_{i\cdot}, \text{Post}_t, X_i, \alpha_i, \lambda_t] = \exp\Big(\alpha_i + \lambda_t + \mathbf{1}\{\text{Anexo} = 6\} \cdot \mathbf{1}\{t > 201712\} + \theta' X_i\Big)$$

# Environment

**Production technology**

$$R(e, L) = A_i\left(e + \delta_i \frac{L^{1-\nu}}{1-\nu}\right), \qquad \nu \in (0, 1), \tag{3}$$

where:

- $A_i > 0$ and $\delta_i > 0$ are firm-specific productivity and relative productivity of hired labor, respectively
- Choice variables: $e \geq 0$ and $L \geq 0$ are owner effort and hired labor, respectively

**Inputs' Costs**

- Labor cost: market wage $(w)$ + fixed costs $F(L) = f \cdot \mathbf{1}[L > 0]$
- Effort cost: $v(e) = \frac{1}{1+\eta} e^{1+\eta}, \qquad \eta \geq 1$

# Environment

**Reporting costs**

$$c(R-\tilde{R}) = \frac{\kappa}{2}\left(\frac{R-\tilde{R}}{R}\right)^2 R, \tag{4}$$

where:

- Reported revenues as a choice variable
- Simplifying assumption: evasion share constant across firm size

**Tax environment**

- Dividend taxes: $\tau_D = 0$
- **Revenue taxes**: $\tau(\tilde{R}, \mathrm{FR}) = \begin{cases} \tau_V(\tilde{R}) & \text{if } \mathrm{FR} < 0.28, \\ \tau_{III}(\tilde{R}) & \text{if } \mathrm{FR} \geq 0.28, \end{cases}$

# Environment

**Social Security**

- Contributions: $T_P(w_o) = t_P \cdot w_o$, where $w_o$ is the amount of owner income paid as wages
- Benefits' Valuation: $\theta_i t_P$ (individual specific)

**Tax-motivated income-shifting costs**

$$\frac{1}{1+\mu}\, w_o^{1+\mu}$$

**Net cost of owner wages**

$$\Gamma(w_o;\, \theta_i) = (1-\theta_i)\, t_P\, w_o + \frac{1}{1+\mu}\, w_o^{1+\mu}, \tag{5}$$

with marginal net cost

# Owner problem

The owner of type $(A_i, \delta_i, \theta_i)$ solves:

$$\begin{aligned} \max_{e,\, L,\, \tilde{R},\, w_o} \quad & \underbrace{(1-\tau)\,\tilde{R}}_{\text{after-tax reported rev.}} + \underbrace{(R-\tilde{R})}_{\text{unreported rev.}} - \underbrace{wL}_{\text{wage bill}} - \underbrace{f}_{\text{fixed cost}} \\ & - \underbrace{v(e)}_{\text{effort cost}} - \underbrace{c(R-\tilde{R})}_{\text{evasion cost}} - \underbrace{\Gamma(w_o;\theta_i)}_{\text{pro-labore cost}} \end{aligned} \tag{6}$$

subject to

$$R = A_i\left(e + \delta_i \frac{L^{1-\nu}}{1-\nu}\right) \tag{7}$$

$$\tilde{R} \leq R \tag{8}$$

$$\mathrm{FR} = \frac{w_o + wL}{\tilde{R}} \tag{9}$$

$$L,\ e,\ w_o > 0. \tag{10}$$

For a solo firm ($L = 0$), the $wL$ and $f$ terms vanish and $R = A_i e$.

# Solution

We do this in two steps:

1. Conditional optimization (fix $\tau$) $\to$ Pre-policy
2. Responses to Factor R rule.

# Conditional Optimization

FOC for reported revenues (taking true revenues as given):

$$\frac{R^* - \tilde{R}^*}{R^*} = \frac{\tau}{\kappa}. \tag{11}$$

- Note that an interior solution requires $\kappa > \tau$
- Very useful expression as $\tilde{R}^* = (1 - \tau/\kappa)R^*$, so we can find the effective tax rate as:

$$\frac{d\Pi^R}{dR} = 1 - \tau + \frac{\tau^2}{2\kappa} \equiv 1 - \underbrace{\tau_{\text{eff}}}_{<\tau}, \tag{12}$$

- Evasion increases the net-of-tax ($s_{\text{eff}} \equiv 1 - \tau_{\text{eff}}$) relative to no evasion
- Under perfect enforcement ($\kappa \to \infty$), $\tau_{\text{eff}} \to \tau$ and $s_{\text{eff}} \to 1 - \tau$

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# Structural elasticities w.r.t statutory net-of-revenue tax

| Margin | Variable | Elasticity w.r.t. $s = 1 - \tau$ | Governed by | Perf. enf. |
|---|---|---|---|---|
| Effort | $e$ | $\frac{1}{\eta} \cdot \frac{s}{s_{\text{eff}}} \cdot \rho(\tau)$; $\rho(\tau) = 1 - \frac{\tau}{\kappa}$ | $\eta$ | $1/\eta$ |
| Labour (int.) | $L$ | $\frac{1}{\nu} \cdot \frac{s}{s_{\text{eff}}} \cdot \rho(\tau)$ | $\nu$ | $1/\nu$ |
| True rev. (solo) | $R$ | $= \varepsilon_{e,s}$ | $\eta$ | $1/\eta$ |
| True rev. (hire) | $R$ | $\omega_e \varepsilon_{e,s} + \omega_L \varepsilon_{L,s}$ | $\eta, \nu$ | Weighted avg. |
| Reported rev. | $\tilde{R}$ | $\varepsilon_{R,s} + s/(\kappa - \tau)$ | $\eta, \nu, \kappa$ | $= \varepsilon_{R,s}$ |
| Pro-labore | $w_o$ | 0 (pre-policy) | — | 0 |
| Hiring rate | $\Pr(L > 0)$ | Distributional (hazard at $\bar{z}$) | $\nu, f, H(\cdot)$ | — |

- Low $\kappa \rightarrow$ large reporting responses $\rightarrow$ smaller real elasticities
- Reported revenue elasticity depends on a real and a mechanical reporting component
- Owner wages do not change with revenue tax rate

# Introducing the Factor R Rule

Pre-policy, all firms face tax rate $\tau_B$. Then, the change in the effective net-of-tax rate is

$$\Delta s_{\text{eff}} = s_{\text{eff}}^A - s_{\text{eff}}^B > 0 \iff \tau_A < \tau_B$$

Pre-policy, firms' optimal Factor R is:

$$\mathrm{FR}_0 = \frac{w_o^* + w \overbrace{(z_i^B)^{1/\nu}}^{L_B^*}}{\underbrace{\rho_B R_B^*}_{\tilde{R_B^*}}},$$

Where $z_i^B = \frac{s_{\text{eff}}^B A_i \delta_i}{w}$ is the productivity-wage ratio, which is distorted by revenue taxation:

$$z_i^A \equiv \frac{s_{\text{eff}}^A A_i \delta_i}{w} = z_i^B \cdot \frac{s_{\text{eff}}^A}{s_{\text{eff}}^B}$$

# Firms above .28 pre-policy

Owner/Firm with primitives $(A_i, \delta_i, \theta_i)$ with optimal choices $\left(\tilde{R}^B, R(e^B, L^B), w_o^B\right)$ satisfying:

$$\mathrm{FR}_0 = \frac{w_o^* + w \overbrace{(z_i^B)^{1/\nu}}^{L_B^*}}{\underbrace{\rho_B \, R_B^*}_{\tilde{R}_B^*}} > .28$$

**What does it happen to them when the Factor R rule is introduced?**

They choose $\left(\tilde{R}^A, R(e^A, L^A), w_o^A\right)$

- Unconstrained case
- Constrained case

# Firms above .28 pre-policy

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# Firms above .28 pre-policy

**Unconstrained case (H1)**

$$\mathrm{FR}_A = \frac{w_o^* + w\,(z_i^A)^{1/\nu}}{\rho_A\, R_B^*} \geq .28$$

**Response to the policy in %**

$$\textit{Effort (\%):} \qquad \frac{\Delta e^{H1}}{e_B^*} = \left(\frac{s_{\text{eff}}^A}{s_{\text{eff}}^B}\right)^{1/\eta} - 1 \approx \frac{1}{\eta} \cdot \frac{\Delta s_{\text{eff}}}{s_{\text{eff}}^B}.$$

$$\textit{Hired labour (\%):} \qquad \frac{\Delta L^{H1}}{L_B^*} = \left(\frac{s_{\text{eff}}^A}{s_{\text{eff}}^B}\right)^{1/\nu} - 1 \approx \frac{1}{\nu} \cdot \frac{\Delta s_{\text{eff}}}{s_{\text{eff}}^B}.$$

$$\textit{Reported revenue (\%):} \qquad \frac{\Delta \tilde{R}^{H1}}{\tilde{R}_B^*} = \underbrace{\frac{\rho_A}{\rho_B} \cdot \frac{\Delta R^{H1}}{R_B^*}}_{\text{real channel (scaled)}} + \underbrace{\frac{\Delta \tau}{\kappa\, \rho_B}}_{\text{reporting channel}}.$$

# Firms above .28 pre-policy

**Constrained case (H2)**

$$\mathrm{FR}_A = \frac{w_o^* + w\,(z_i^A)^{1/\nu}}{\rho_A\, R_B^*} < .28$$

Since this optimum is not feasible (it implies $\tau_B$ which leads to $FR_0 > .28$), the unique equilibrium is the *contrained* solution

- $w_o$ is now use to satisfy $FR = .28$
- The owner solves the same problem with $\tau_A$ subject to $FR = .28$

**Augmented tax rate**: From the reported revenue ($\tilde{R}$) FOC, note:

$$\kappa \frac{R - \tilde{R}}{R} = \underbrace{\tau_A + 0.28\,\Gamma'(w_o^c;\theta_i)}_{\tilde{\tau}_A}$$

# Constrained case (H2)

The constrained equilibrium satisfies:

$$
\begin{aligned}
w_o^c &= 0.28\,\rho^c\,R^c - w\,L^c, && (13)\\
\tilde{\tau}_A &= \tau_A + 0.28[(1-\theta_i)t_P + (w_o^c)^\mu]\,, && (14)\\
s_{\mathrm{eff}}^c &= 1 - \tilde{\tau}_A + (\tilde{\tau}_A)^2/(2\kappa), \quad \rho^c = 1 - \tilde{\tau}_A/\kappa, && (15)\\
e^c &= (s_{\mathrm{eff}}^c A_i)^{1/\eta}, && (16)\\
L^c &= \left(\frac{s_{\mathrm{eff}}^c A_i \delta_i}{w_{\mathrm{eff}}^c}\right)^{1/\nu}, \quad w_{\mathrm{eff}}^c = w[1-(1-\theta_i)t_P - (w_o^c)^\mu]\,, && (17)\\
R^c &= A_i\left(e^c + \delta_i \frac{(L^c)^{1-\nu}}{1-\nu}\right). && (18)
\end{aligned}
$$

1. Note that $w_o^c = F(w_o^c)$. Since $\Gamma'' > 0$, F is a contraction $\rightarrow$ Unique solution.
2. Isomorphic to H1 problem with augmented tax rate $\tilde{\tau}_A$ and wage $w_{\mathrm{eff}}^c$

# Contrained Case (H2)

Change in owner wages:

$$\Delta w_o^{H2} = \underbrace{-\tilde{R}_B^*(\mathrm{FR}_0 - 0.28)}_{\text{(a) pre-policy buffer}} + \underbrace{0.28\,(\rho^c R^c - \rho_B R_B^*)}_{\text{(b) reported rev. change}} - \underbrace{w\,\Delta L^{H2}}_{\text{(c) wage bill offset}} \quad . \tag{19}$$

# Firms below .28 pre-policy

Owner/Firm with primitives $(A_i, \delta_i, \theta_i)$ with optimal choices $\left(\tilde{R}^B, R(e^B, L^B), w_o^B\right)$ satisfying:

$$\mathrm{FR}_0 = \frac{w_o^* + w \overbrace{\left(z_i^B\right)^{1/\nu}}^{L_B^*}}{\underbrace{\rho_B \, R_B^*}_{\tilde{R}_B^*}} < .28$$

**What does it happen to them when the Factor R rule is introduced?**
They choose $\left(\tilde{R}^A, R(e^A, L^A), w_o^A\right)$

- $U_B^*$: stay below 0.28, face $\tau_B$ (pre-policy unchanged).
- $U_A^{\mathrm{unc}}$: unconstrained optimum at $\tau_A$ (feasible only if resulting $\mathrm{FR} \geq 0.28$).
- $U^c$: constrained optimum with $\mathrm{FR} = 0.28$ binding at $\tau_A$.

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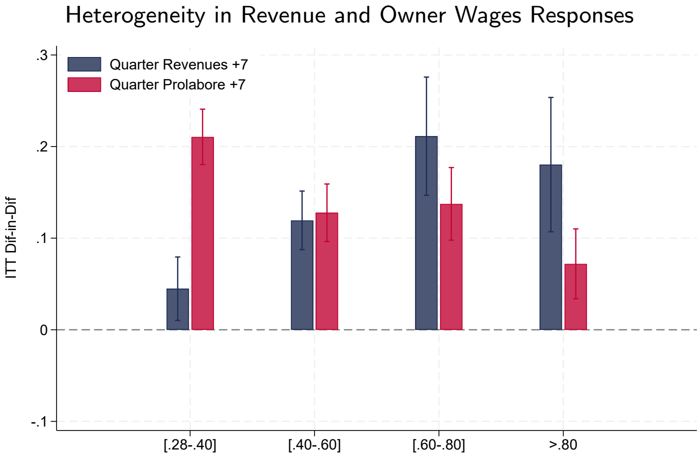
# Firms below .28 pre-policy

- **L1**: $U_A^{\mathrm{unc}} \geq U^c$ and $U_A^{\mathrm{unc}} > U_B^*$. Unconstrained $\tau_A$ optimum has $\mathrm{FR} \geq 0.28$; firm crosses unforced.
- **L2**: $U_A^{\mathrm{unc}} < U^c$ (unconstrained $\tau_A$ has $\mathrm{FR} < 0.28$) but $U^c > U_B^*$. Firm bunches at $\mathrm{FR} = 0.28$.
- **L3**: $U^c \leq U_B^*$. Firm does not respond.

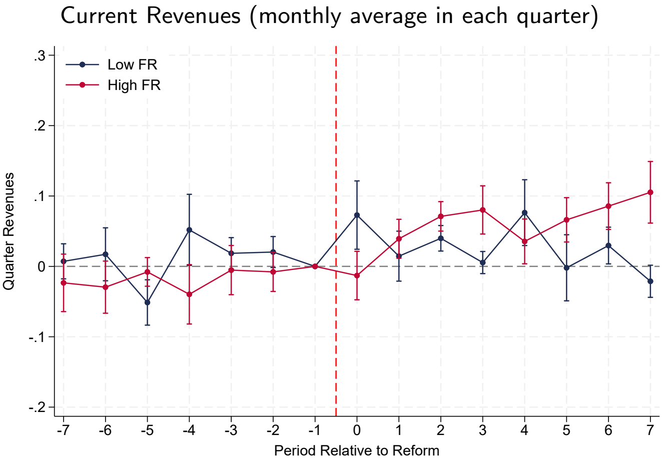
# Firm FR > .28 - Heterogeneity




**Revenue responses are larger in very high FR and prolabore closer to the threshold**

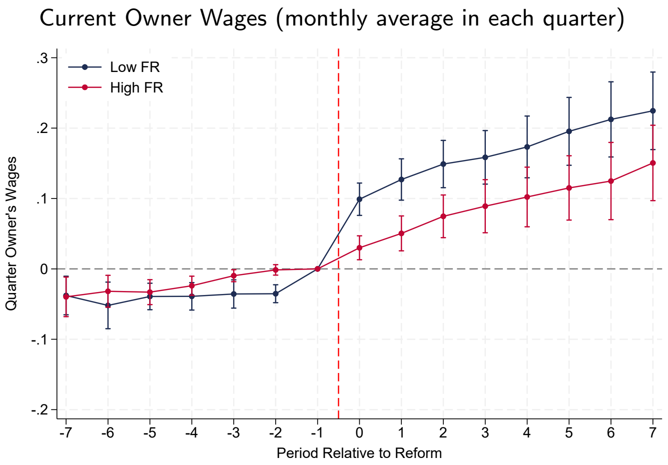
Back Revenues Back Prolabore

# Firm FR > .28 - Heterogeneity




Back Revenues

# Firm FR > .28 - Heterogeneity




Back

# Firm FR > .28 - Heterogeneity

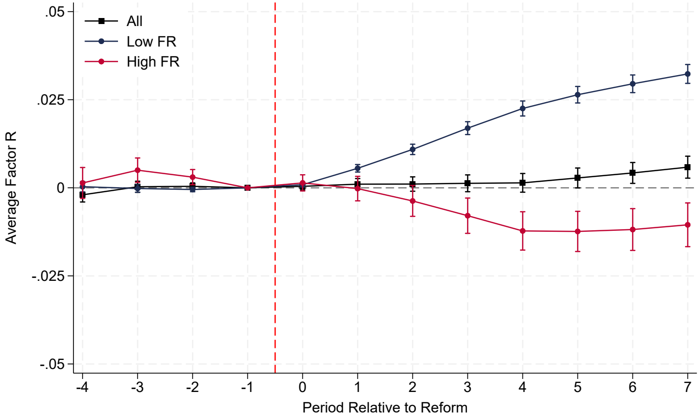

Back

## Descriptive Statistics: Firms in 2017-Q4

| | Accumulated 12-month | | | | Current (Quarter) | | | |
|---|---|---|---|---|---|---|---|---|
| | Low FR | | High FR | | Low FR | | High FR | |
| | Treated | Control | Treated | Control | Treated | Control | Treated | Control |
| Revenues | 308,565 | 543,037 | 220,681 | 513,755 | 28,403 | 51,261 | 20,575 | 47,432 |
| Owner Wages | 17,526 | 17,673 | 33,878 | 29,774 | 1,558 | 1,558 | 2,857 | 2,475 |
| Employees' Payroll | 23,927 | 38,690 | 70,277 | 220,735 | 2,564 | 4,145 | 7,226 | 22,383 |
| $\mathbf{1}$[Revenues = 0] | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $\mathbf{1}$[Owner Wages = 0] | 0.278 | 0.347 | 0.092 | 0.128 | 0.298 | 0.371 | 0.105 | 0.150 |
| $\mathbf{1}$[Employees' Payroll = 0] | 0.581 | 0.476 | 0.269 | 0.100 | 0.603 | 0.512 | 0.289 | 0.122 |
| $\mathbf{1}$[Total Payroll = 0] | 0.182 | 0.221 | 0.000 | 0.000 | 0.198 | 0.244 | 0.003 | 0.004 |
| N Firms | 19,863 | 29,594 | 16,320 | 17,692 | 19,863 | 29,594 | 16,320 | 17,692 |

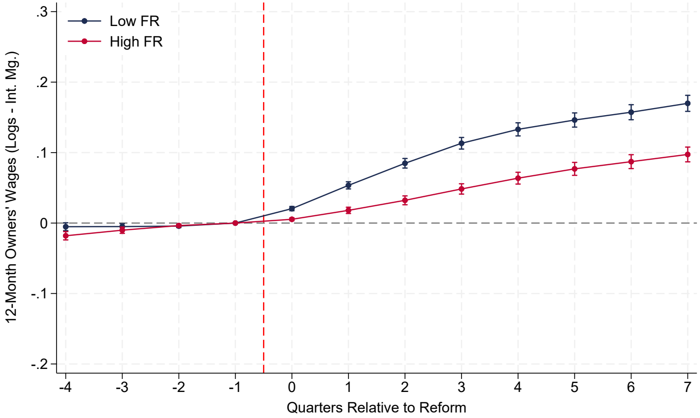
US$ 1 $\simeq$ R$3.34 (Dec–2017) Back

# Effect on Average Factor R




Back

# Log Specification for 12-month Owner Wages

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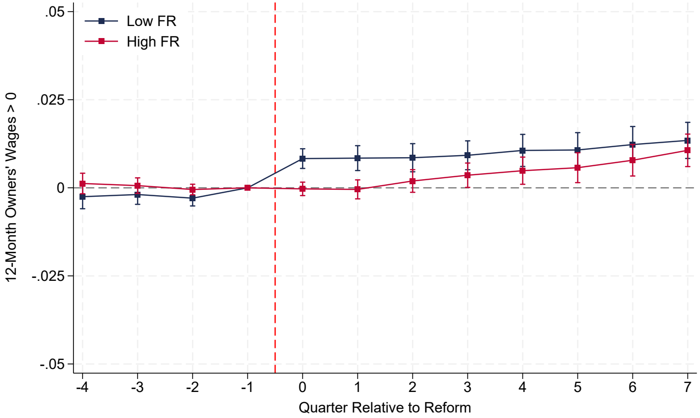
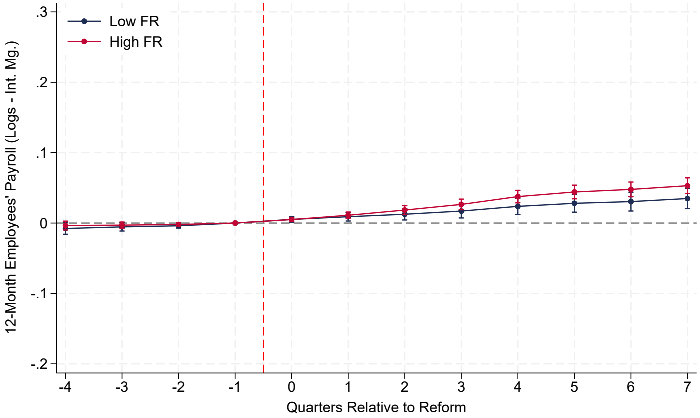
# Positive 12-month Owner Wages




Back

# Log Specification for 12-month Employees' Payroll

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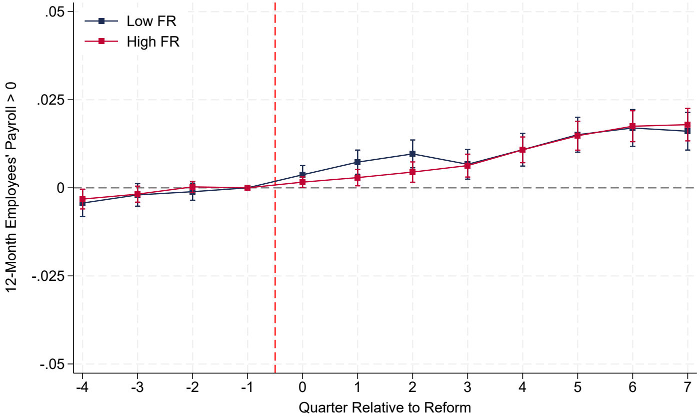


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# Firms above 28% pre-policy

**What we are identifying:**

$$\varepsilon_R = \frac{\Delta R}{\Delta(1-\tau_R)} \frac{(1-\tau_R)}{R} = \pi_A \varepsilon_A + \pi_{con} \varepsilon_{con}$$

**Estimation** (sample conditioned on pre-policy Factor R $>$ .28)

$$\begin{aligned} \Delta_{-1,t} \ln y_i &= \alpha_0 + \varepsilon_{R,t} \Delta_{-1,t} \ln (1-\tau_i) + u_{it} \\ \Delta_{-1,t} \ln (1-\tau_i) &= \gamma_0 + \gamma_1 \mathbf{1}[\text{Anexo VI} = 1] + v_{it} \end{aligned} \tag{20}$$

- Not well-documented in the literature

→ Lobel et al. 2024: Introduction of minimum revenue tax shielded $\varepsilon_R = .4$

→ Heim 2010: ETI for self-employed in the intensive margin $\varepsilon_R = .9$ (half real effects)

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# Revenue Elasticitiy

Revenue Elasticity Estimates Over Time Horizons

| | 12-month Revenue | | Quarter Revenue | |
|---|---|---|---|---|
| | 1 Year | 2 Year | 1 Year | 2 Year |
| $\varepsilon_R$ | .392 | .887 | .994 | 1.374 |
| SE | (0.045) | (0.063) | (0.067) | (0.084) |
| CI | [0.30, 0.48] | [0.76, 1.01] | [0.86, 1.13] | [1.21, 1.54] |
| Reduced Form | .040 | .089 | 0.101 | 0.137 |
| First Stage ($\gamma_1$) | 0.101 | 0.100 | 0.101 | 0.100 |
| Observations | 34,014 | 34,014 | 34,014 | 34,014 |

Back

# Firms below 28%: Income-Shifting Elasticity

$$\varepsilon_S = \frac{\Delta S}{\Delta\left(\frac{P^W}{P^D}\right)} \frac{\left(\frac{P^W}{P^D}\right)}{S} \simeq \frac{\Delta \ln S}{\Delta \ln\left(\frac{P^W}{P^D}\right)} \tag{21}$$

- S is the sh. of total owner income paid as wages
- $\frac{P^W}{P^D}$ is the relative return of reporting as wages relative to dividends

**How do we estimate the change in incentives in non-linear schemes?**

- Key concept: change in the return of reporting one unit of income as wages relative to dividends

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# Basic Case: No Personal Income Tax

**Return per dollar of owner wages reported**

- $\rightarrow$ Revenues: $R_j$
- $\rightarrow$ Total Payroll (incl. owner wages) before the policy: $P_{0j}$
- $\rightarrow$ Factor R: $FR_j = \frac{P_{0j}}{R_j}$
- $\rightarrow$ Threshold: $T = 0.28$

$$\Delta W^*_{Oj} = (T - FR_j)R_j \qquad \rightarrow \textbf{Increase in owner wages to get the revenue tax cut}$$

$$\text{Saving}_j = \Delta\tau_R \cdot R_j \qquad \rightarrow \textbf{Tax savings from getting to the threshold}$$

Then, the return per dollar of additional owner wages used to cross the threshold is:

$$\frac{\text{Saving}_j}{\Delta W^*_{Oj}} = \frac{\Delta\tau_R \cdot R_j}{(T - FR_j)R_j} = \frac{\Delta\tau_R}{(T - FR_j)} \tag{22}$$

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## Basic Case: No Personal Income Tax

Then, the change in returns of reporting one unit of income as wages relative to dividends is:

$$\Delta \ln\left(\frac{P_j^W}{P_j^D}\right) = \ln\left(\frac{1+\frac{\Delta\tau_R}{(T-FR_j)}}{1-\tau_D}\right) - \ln\left(\frac{1}{1-\tau_D}\right) \tag{23}$$

In the absence of dividend taxes ($\tau_D$) this expression reduces to:

$$\Delta \ln\left(\frac{P_j^W}{P_j^D}\right) = \ln\left(1+\frac{\Delta\tau_R}{(T-FR_j)}\right) \tag{24}$$

Defining $s_j$ **as the sh. of owner income paid as wages**, the income-shifting elasticity is:

$$\varepsilon_S = \frac{1}{N}\sum_j \varepsilon_{Sj} = \frac{1}{N}\sum_j \left(\frac{\Delta \ln(s_j)}{\ln\left(1+\frac{\Delta\tau_R}{(T-FR_j)}\right)}\right) \tag{25}$$

# Basic Case: No Personal Income Tax

**What we are identifying:**

$$\varepsilon_S = \frac{\Delta S}{\Delta\left(\frac{P^W}{P^D}\right)} \frac{\left(\frac{P^W}{P^D}\right)}{S} \simeq \frac{\Delta \ln S}{\Delta \ln\left(\frac{P^W}{P^D}\right)} = \pi_3 \varepsilon_{S3}$$

**Estimation thourgh 2SLS** (sample conditioned on pre-policy Factor R $<$ .28)

$$\begin{aligned} \Delta_{-1,t} \ln S_i &= \alpha_0 + \varepsilon_{S,t} \ln\left(1 + \frac{(\Delta_{-1,t}\tau_R)_i}{(T - FR_i)}\right) + u_{it} \\ \ln\left(1 + \frac{(\Delta_{-1,t}\tau_R)_i}{(T - FR_i)}\right) &= \gamma_0 + \gamma_1 \mathbf{1}[\text{Anexo VI} = 1] + v_{it} \end{aligned} \tag{26}$$

# Income-Shifting Elasticitiy

Income-Shifting Elasticity Estimates Over Time Horizons

| | 12-month Prolabore | | Quarter Prolabore | |
|---|---|---|---|---|
| | 1 Year | 2 Year | 1 Year | 2 Year |
| $\varepsilon_S$ | .392 | .887 | .994 | 1.374 |
| SE | (0.045) | (0.063) | (0.067) | (0.084) |
| CI | [0.30, 0.48] | [0.76, 1.01] | [0.86, 1.13] | [1.21, 1.54] |
| Reduced Form | .040 | .089 | 0.101 | 0.137 |
| First Stage ($\gamma_1$) | 0.597 | 0.583 | 0.597 | 0.583 |
| Observations | 49,457 | 49,457 | 49,457 | 49,457 |

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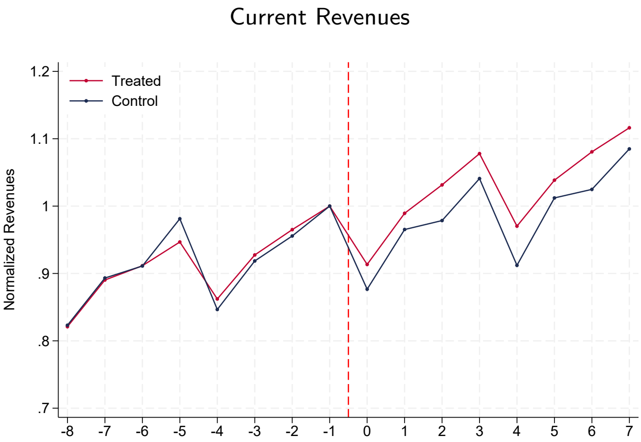
# From the Firm to the Individual Tax Data

Dif-in-Dif Coefficients (Single-owned firms with DIRPF information)

| | Firm | | | Owner | | |
|---|---|---|---|---|---|---|
| | Prolabore | Revenues | Rev. Tax | Taxable Inc. | Dividends | IRPF |
| $2018 \cdot \mathbf{1}[\text{Treated} = 1]$ | .122 | .017 | -.157 | .025 | -.192 | -.012 |
| | [.07, .17] | [-.20, -.12] | [.00, .05] | [1.21, 1.54] | [-.31, -.08] | [-.07, .04] |
| $2019 \cdot \mathbf{1}[\text{Treated} = 1]$ | .178 | -.018 | -.263 | .033 | -.258 | -.023 |
| | [0.12, 0.23] | [-.07, .03] | [-.31, -21] | [.01, .06] | [-.40, -.11] | [-.09, .04] |
| $2016 \cdot \mathbf{1}[\text{Treated} = 1]$ | -.042 | -.027 | -.045 | -.00 | .017 | .006 |
| | [-.08, -.00] | [-.09, -.00] | [-.09, .00] | [-.02, .02] | [-.10, .14] | [-.05, .06] |
| Observations | 20,744 | 30,108 | 30,108 | 27,380 | 5,476 | 15,560 |

# Treated vs. Control (All)



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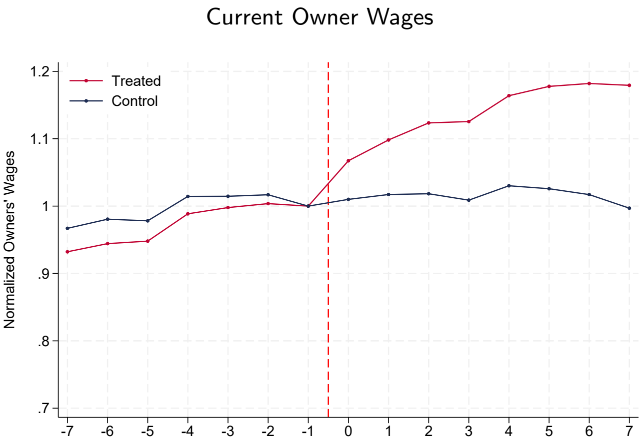

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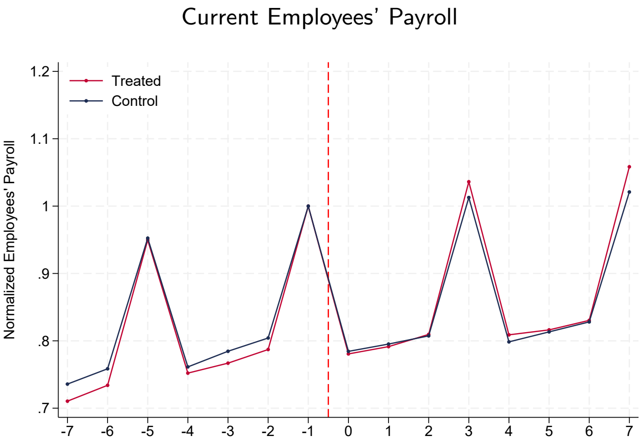

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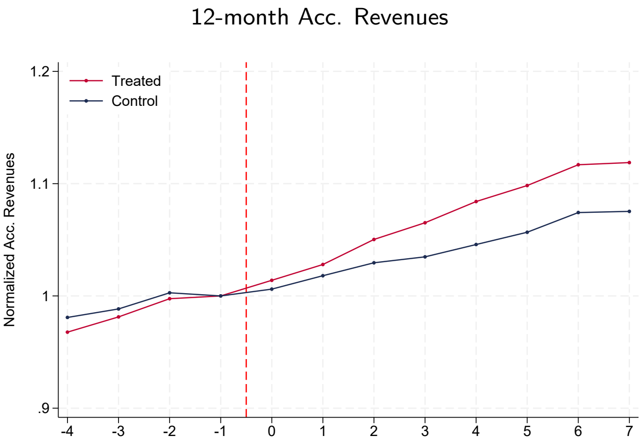

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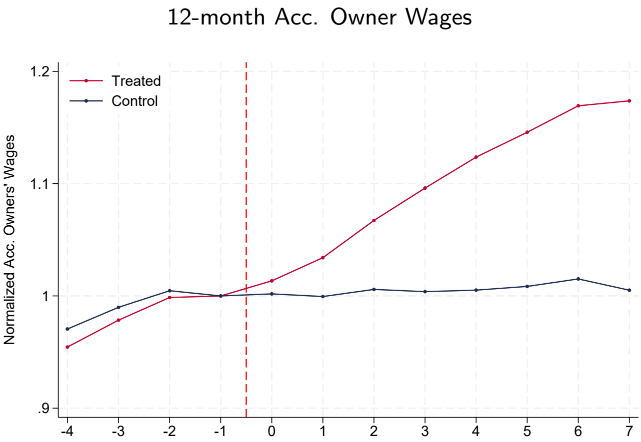
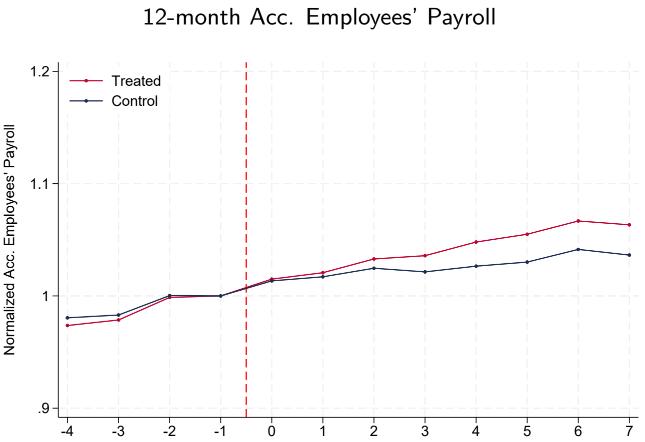
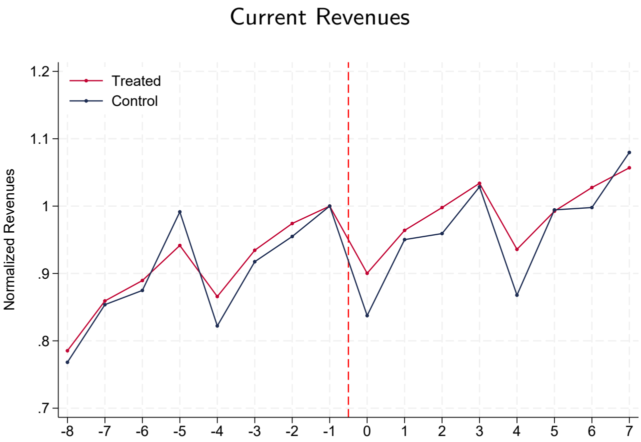


# Treated vs. Control (All)



# Treated vs. Control (Low)



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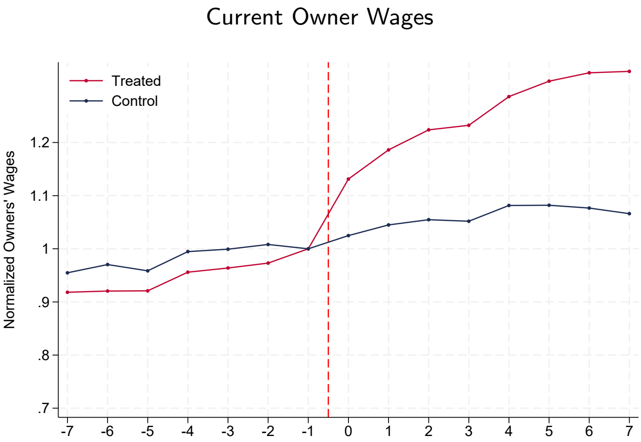
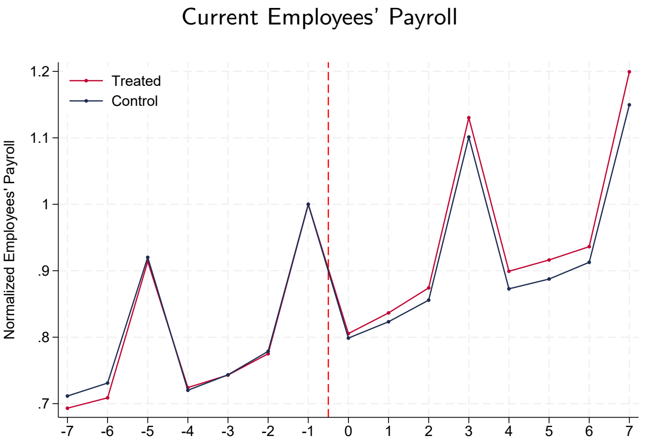
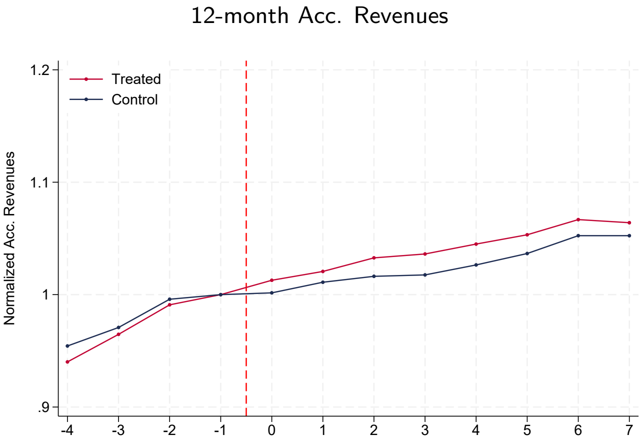


# Treated vs. Control (Low)



# Treated vs. Control (Low)

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# Treated vs. Control (Low)


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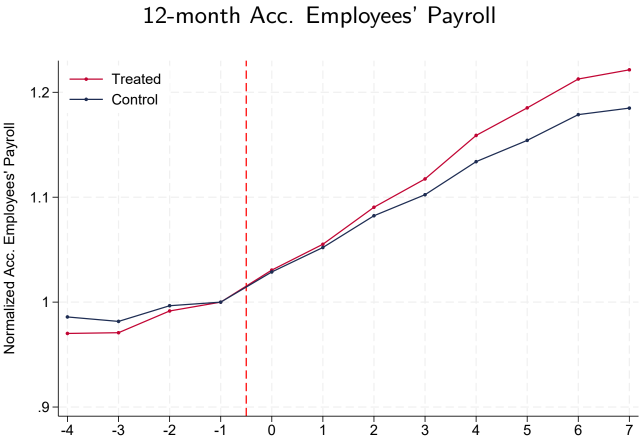
# Treated vs. Control (Low)

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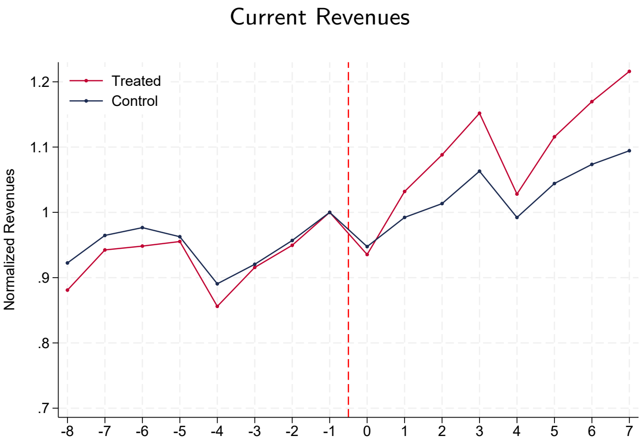

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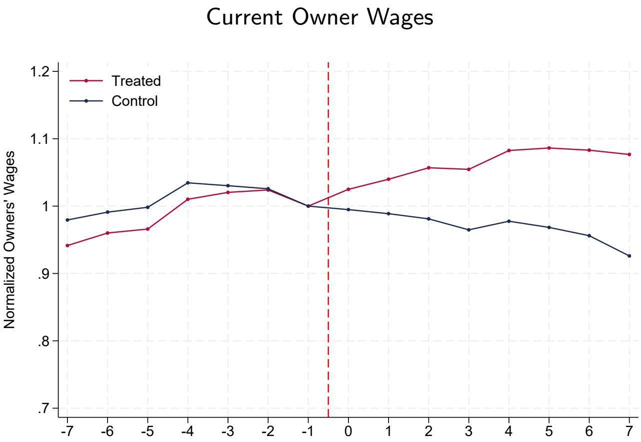

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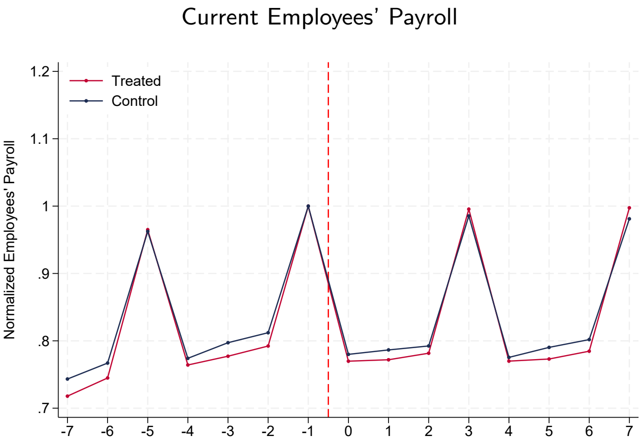

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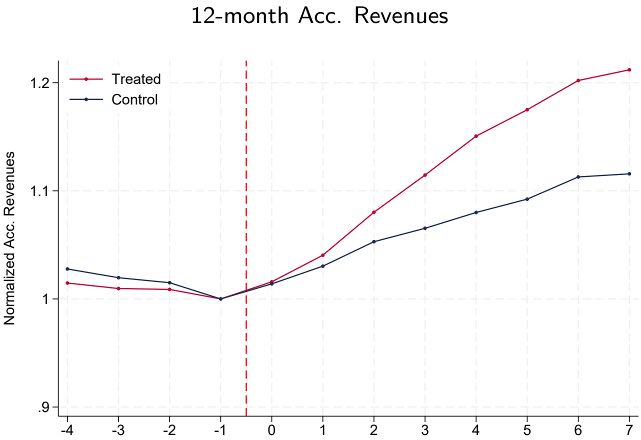


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# Treated vs. Control (High)


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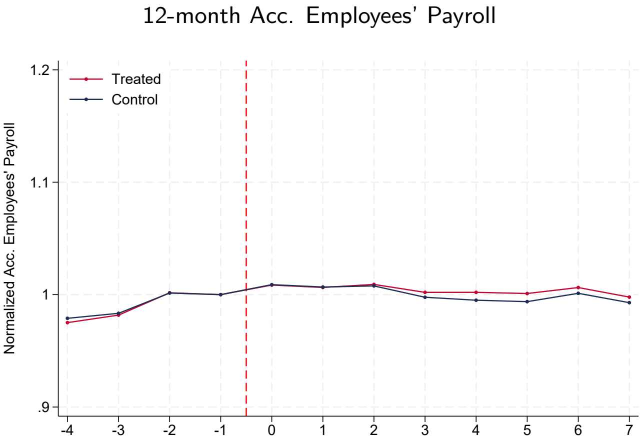
# Treated vs. Control (High)

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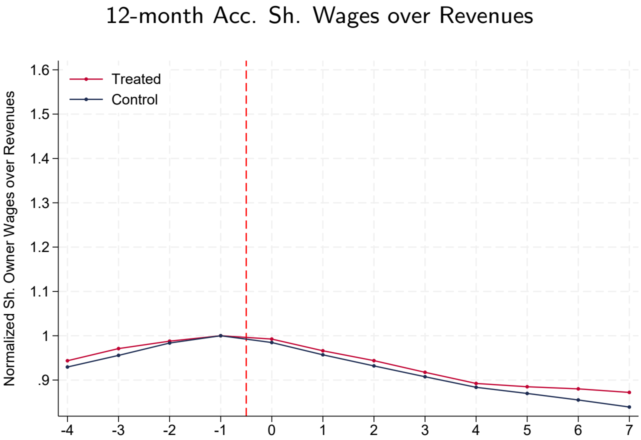


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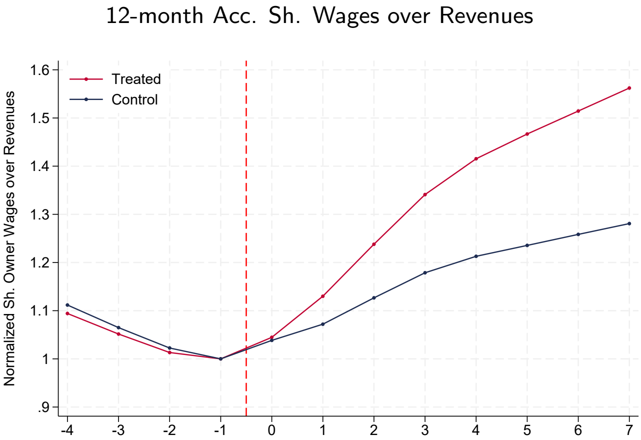


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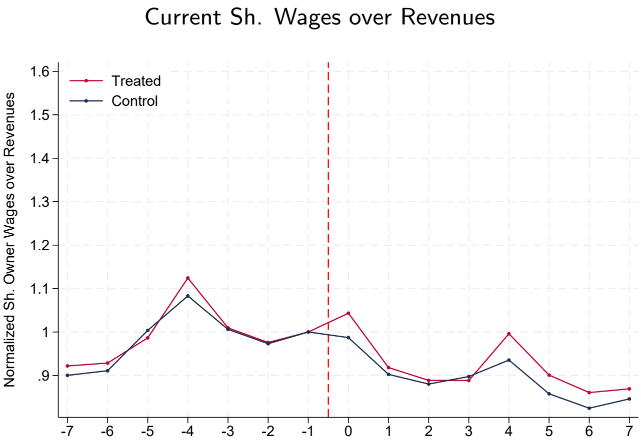
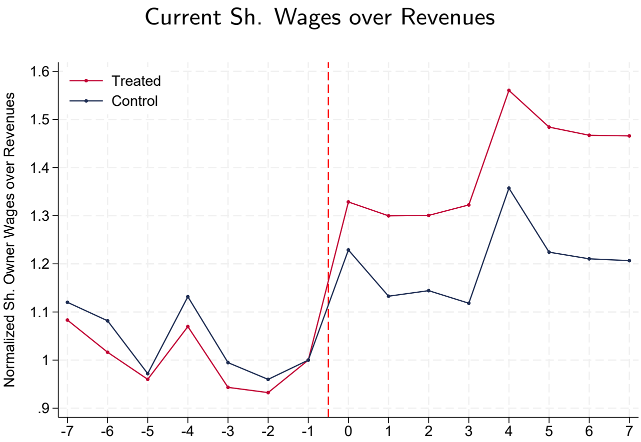
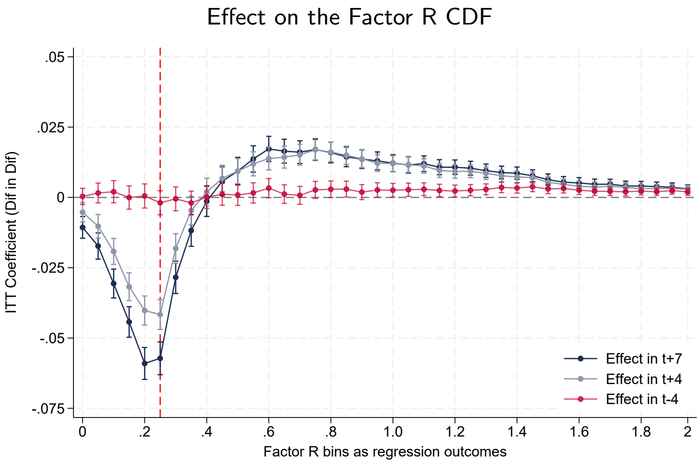


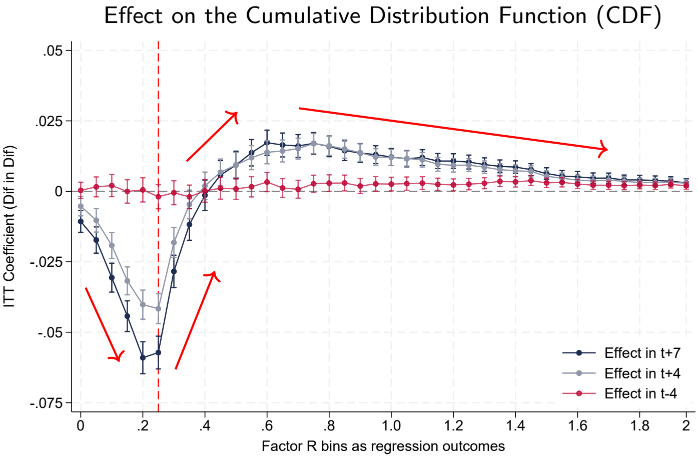
# Treated vs. Control (Low)



# Effects on the location in the Factor R distribution

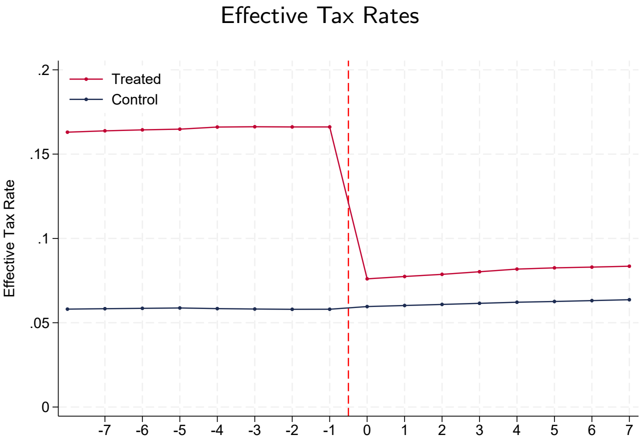


# Effects on the location in the Factor R distribution




**84% of the bunching comes from below the threshold, and 16% comes from above** Avg. FR

# Treated vs. Control (High)

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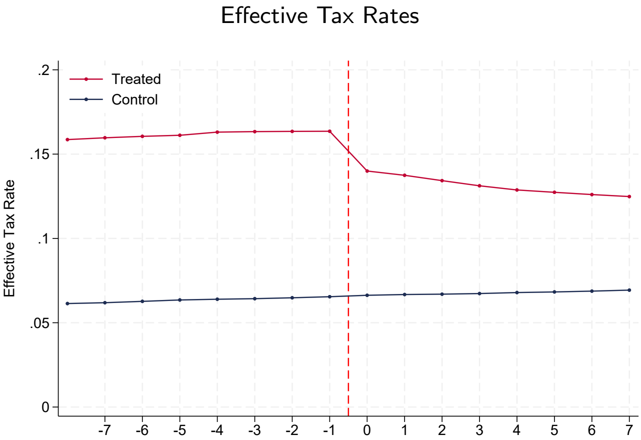
# Treated vs. Control (High)

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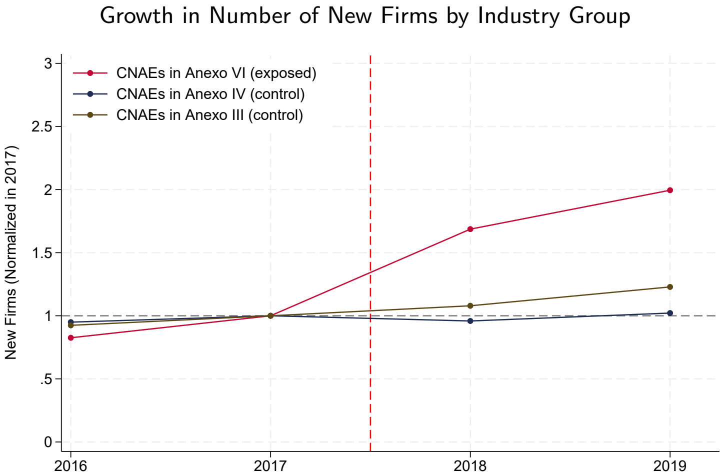
# Relabeling industry code?




**Little evidence of a compensating drop in other Service industries** Back Anexo III